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NAME: _____

VCE® SPECIALIST MATHEMATICS

Units 3 & 4 Practice Written Examination 2

Reading time: 15 minutes
Writing time: 2 hours

QUESTION AND ANSWER BOOK

Structure of book

Section	*Number of questions*	*Number of questions to be answered*	*Number of marks*
A	20	20	20
B	6	6	60
			Total 80

- Students are permitted to bring into the examination room: pens, pencils, highlighters, erasers, sharpeners, rulers, a protractor, set-squares, aids for curve sketching, one bound reference, one approved technology (calculator or software) and, if desired, one scientific calculator. Calculator memory DOES NOT need to be cleared. For approved computer-based CAS, full functionality may be used.
- Students are NOT permitted to bring into the examination room: blank sheets of paper and/or correction fluid/tape.

Materials supplied

- Question and answer book of 30 pages.
- Formula sheet.
- Answer sheet for multiple-choice questions.

Instructions

- Write your **name** in the space provided above on this page.
- Write your **name** on the multiple-choice answer sheet.
- Unless otherwise indicated, the diagrams are **not** drawn to scale.
- All written responses must be in English.

Students are NOT permitted to bring mobile phones and/or any other unauthorised electronic devices into the examination room.

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SECTION A – Multiple-Choice Questions

Instructions for Section A

Answer **all** questions in pencil on the answer sheet provided for Multiple-Choice Questions.
Choose the response that is **correct** for the question.
A correct answer scores 1; an incorrect answer scores 0.
Marks will **not** be deducted for incorrect answers.
No marks will be given if more than one answer is completed for any question.
Unless otherwise indicated, the diagrams in this book are **not** drawn to scale.
Take the **acceleration due to gravity** to have magnitude g ms^{-2}, where $g = 9.8$.

Question 1

Consider the following statements:

P: $n^2 - 6n + 5$ is even. Q: n is odd.

Which one of the following statements would be assumed in order to prove the conjecture

If $n^2 - 6n + 5$ is even then n is odd

using proof by contradiction?

A. $\neg P \Rightarrow \neg Q$

B. $\neg P \wedge \neg Q$

C. $P \wedge \neg Q$

D. $\neg P \wedge Q$

Question 2

Two of the asymptotes of the graph of $y = \frac{|x|^3 - 1}{ax^2 + b|x| + c}$ are given by $y = -2x + 14$ and $x = 6$.

Possible values of *a*, *b* and *c* are, respectively

A. $-\frac{1}{2}, -\frac{7}{2}, -39$

B. $\frac{1}{2}, -\frac{7}{2}, 3$

C. $\frac{1}{2}, \frac{7}{2}, -3$

D. $-2, \frac{7}{2}, 3$

Question 3

The graph of $y = a - b^2 \arctan\left(\frac{x}{2}\right)$, where $a, b \in R \setminus \{0\}$, will have no x-intercept only when

A. $b^2 \le \left|\frac{2a}{\pi}\right|$

B. $b^2 \ge \left|\frac{2a}{\pi}\right|$

C. $0 < b^2 \le \frac{2a}{\pi}$ and $a > 0$

D. $0 < b^2 < \frac{2a}{\pi}$ and $a > 0$

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Question 4

The complex numbers z, $-iz$ and $z - iz$, where $z \neq 0$, are plotted in the Argand plane to form the vertices of a triangle.

The area of this triangle is given by

A. $\frac{1}{2}\left(|z|^2 + |z|\right)$

B. $\frac{\sqrt{3}}{2}|z|$

C. $\frac{\sqrt{3}}{2}|z|^2$

D. $\frac{1}{2}|z|^2$

Question 5

If $\text{Arg}(z+1) = \frac{\pi}{3}$ and $\text{Arg}(z-1) = \frac{\pi}{2}$, then $\text{Arg}(z-4)$ is equal to

A. $\pi - \arctan\left(\frac{\sqrt{3}}{2}\right)$

B. $-\arctan\left(\frac{\sqrt{3}}{2}\right)$

C. $\pi - \arctan\left(\frac{2\sqrt{3}}{3}\right)$

D. $-\arctan\left(\frac{2\sqrt{3}}{3}\right)$

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Question 6

The relation $|z+a| - |z-3a| = k$, where $z \in C$ and $a \in R \setminus \{0\}$, will define a ray when

A. $0 < k \le 3a$

B. $a \le k \le 3a$

C. $k = 3a$

D. $k = 4a$

Question 7

Consider the following pseudocode.

```
define f(t,x)
     return (t^2+1)/(x^3+1)
a ← ___
b ← 4
n ← 10
h ← (b-a)/n
t ← a
x ← 3
for i from 1 to n
     x ← x + h*f(t,x)
     t ← t+h
end for
print x to three decimal places
```

For which one of the following values of a will the pseudocode print `3.504`?

A. 2.1

B. 2.2

C. 2.3

D. 2.4

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Question 8

A tank initially contains 16 kg of salt that is dissolved in 90 L of water. A solution containing 6 kg of salt per litre of water flows into the tank at a rate of 5 L per minute and the mixture in the tank is kept well stirred. At the same time, 7 L of the mixture flows out of the tank per minute.
A differential equation defining the concentration, c kg per litre of water, of salt in the tank at time t minutes, for a non-zero volume of mixture, is

A. $\frac{dc}{dt} = 30 - 7c$

B. $\frac{dc}{dt} = \frac{30 - 5c}{90 - 2t}$

C. $\frac{dc}{dt} = -\frac{7c}{90 - 2t}$

D. $\frac{dc}{dt} = \frac{30 - 7c}{90 - 2t}$

Question 9

The position vector of a particle moving in the Cartesian plane, at time t, is given by

$$\underset{\sim}{r}(t) = \frac{|3 - t^2|}{t+1}\underset{\sim}{i} - \frac{4}{t+2}\underset{\sim}{j}, \text{ where } t \geq 0.$$

What is the slope of the tangent to the path of the particle when $t = 2$?

A. $\frac{9}{44}$

B. $-\frac{9}{44}$

C. $\frac{11}{36}$

D. $-\frac{11}{36}$

Question 10

For n a nonnegative integer, let $I_n = \int_0^{\frac{\pi}{3}} \sec^n(x)\, dx$. If $n \geq 2$, then I_n equals

A. $\frac{2^{n-2}}{n-1}\sqrt{3} - \frac{n-2}{n-1} I_{n-2}$

B. $\frac{2^{n-2}}{n-1}\sqrt{3} + \frac{n-2}{n-1} I_{n-2}$

C. $\frac{2^{n-2}}{n-1}\sqrt{3} + (n-2) I_{n-2}$

D. $(n-1)2^{n-2}\sqrt{3} I_{n-2}$

Question 11

The area of the surface generated by rotating the curve with equation $y = \tan^{-1}(x)$ about the line $x = 1$ from the points where $x = 2$ to $x = 4$ equals

A. $2\pi \int_{\tan^{-1}(2)}^{\tan^{-1}(4)} (\tan(y) - 1)\sqrt{1 + \sec^2(y)}\, dy$

B. $2\pi \int_{\frac{\pi}{4}}^{\tan^{-1}(3)} (\tan(y) + 1)\sqrt{1 + \sec^4(y)}\, dy$

C. $2\pi \int_{\tan^{-1}(2)}^{\tan^{-1}(4)} (\tan(y) - 1)\sqrt{1 + \sec^4(y)}\, dy$

D. $2\pi \int_{\frac{\pi}{4}}^{\tan^{-1}(3)} (\tan(y)+1)\sqrt{1+\sec^2(y)}\ dy$

Question 12

Consider the vectors $\underset{\sim}{a}$ and $\underset{\sim}{b}$ such that $|\underset{\sim}{a}|=4$, $|\underset{\sim}{b}|=3$ and $|\underset{\sim}{a}-\underset{\sim}{b}|=2$.

Then $|\underset{\sim}{a}+\underset{\sim}{b}|$ equals

A. $\sqrt{21}$

B. 5

C. $\sqrt{38}$

D. $\sqrt{46}$

Question 13

Consider the vectors $\underset{\sim}{a}$ and $\underset{\sim}{b}$ such that $\underset{\sim}{a}\cdot\underset{\sim}{b}=4$ and $|\underset{\sim}{a}\times\underset{\sim}{b}|=7$. The angle between $\underset{\sim}{a}$ and $\underset{\sim}{b}$, correct to the nearest degree, is

A. 50

B. 55

C. 60

D. 65

Question 14

A particle starts from rest and moves in a straight line. The velocity, v ms^{-1}, of the particle after t seconds is $v=\tan^{-1}(x)$, where x meters is the distance the particle has moved.

If the particle moves 2 metres in 3 seconds, the time it takes to move 5 metres is approximately

A. 1.52 seconds

B. 2.36 seconds

C. 3.52 seconds

D. 5.36 seconds

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Question 15

An object with an initial velocity of 6 ms^{-1} moves in a straight line with constant acceleration. After ten seconds it is moving at 8 ms^{-1} in the **opposite** direction to its initial velocity.

The distance, in metres, of the object from its starting point after twelve seconds is

A. 28.8

B. 57.6

C. 68.6

D. 86.4

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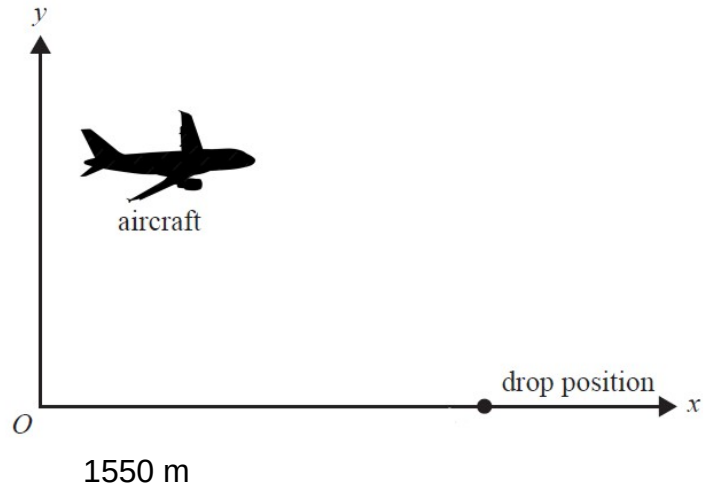
Question 16

A supply aircraft is to release a package so that it lands at a drop position, as shown in the diagram below.



The position vector of the supply aircraft relative to a fixed observation point O is

$\underset{\sim}{r}(t) = 50t\,\underset{\sim}{i} + (600 - 3t)\,\underset{\sim}{j}$, $0 \le t \le 120$ seconds.

The drop position is located 1550 m horizontally from O. $\underset{\sim}{i}$ is a horizontal unit vector in the direction of the drop position and $\underset{\sim}{j}$ is a unit vector vertically up. Displacement components are measured in metres.

When the aircraft is at a height of 540 m, it releases the package. Assuming negligible air resistance acting on the package, how far short of the drop position, correct to the nearest meter, does the package land?

A. 36

B. 40

C. 44

D. 48

Question 17

The planes

$$2x+(k-4)y+(3-k)z=1, \quad 2y+(k-3)z=2, \quad x-2y+z=1$$

intersect along a common line when

A. $k=-1$ and $k=2$

B. $k=2$

C. $k=-1$

D. $k \in R \setminus \{-1, 2\}$

Question 18

If the vector $3\underset{\sim}{i}-7\underset{\sim}{j}+\gamma\underset{\sim}{k}$ is perpendicular to the lines $\frac{x-1}{3}=\frac{3-y}{\alpha}=\frac{z+1}{4}$ and $\frac{x+1}{\beta}=\frac{y-2}{3}=\frac{z-1}{5}$, then

A. $\alpha=1, \beta=-2, \gamma=-4$

B. $\alpha=-1, \beta=-2, \gamma=-\frac{1}{2}$

C. $\alpha=-2, \beta=3, \gamma=\frac{5}{4}$

D. $\alpha=-3, \beta=2, \gamma=3$

Question 19

Consider a random variable X with probability density function

$$f(x) = \begin{cases} 2x, & 0 \le x \le 1 \\ 0, & x < 0 \text{ or } x > 1 \end{cases}.$$

If a large number of samples, each of size 80, is taken from this distribution, then the distribution of sample means, $\overline{X}$, is approximately normal with a standard deviation of

A. $\frac{\sqrt{5}}{90}$

B. $\frac{\sqrt{5}}{80}$

C. $\frac{\sqrt{10}}{120}$

D. $\frac{\sqrt{5}}{72}$

Question 20

The lifespan of a certain type of lightbulb is normally distributed with a mean of μ hours and a standard deviation of 420 hours.

A random sample of these lightbulbs is collected and used to calculate an approximate 90% confidence interval for μ. If the confidence interval is (14 428, 14 572) then the sample size is

A. 92

B. 90

C. 88

D. 86

END OF SECTION A

SECTION B

Instructions for Section B

Answer **all** questions in the spaces provided.
Unless otherwise specified, an **exact** answer is required to a question.
In questions where more than one mark is available, appropriate working **must** be shown.
Unless otherwise indicated, the diagrams in this book are **not** drawn to scale.

Take the **acceleration due to gravity** to have magnitude g ms^{-2}, where $g = 9.8$.

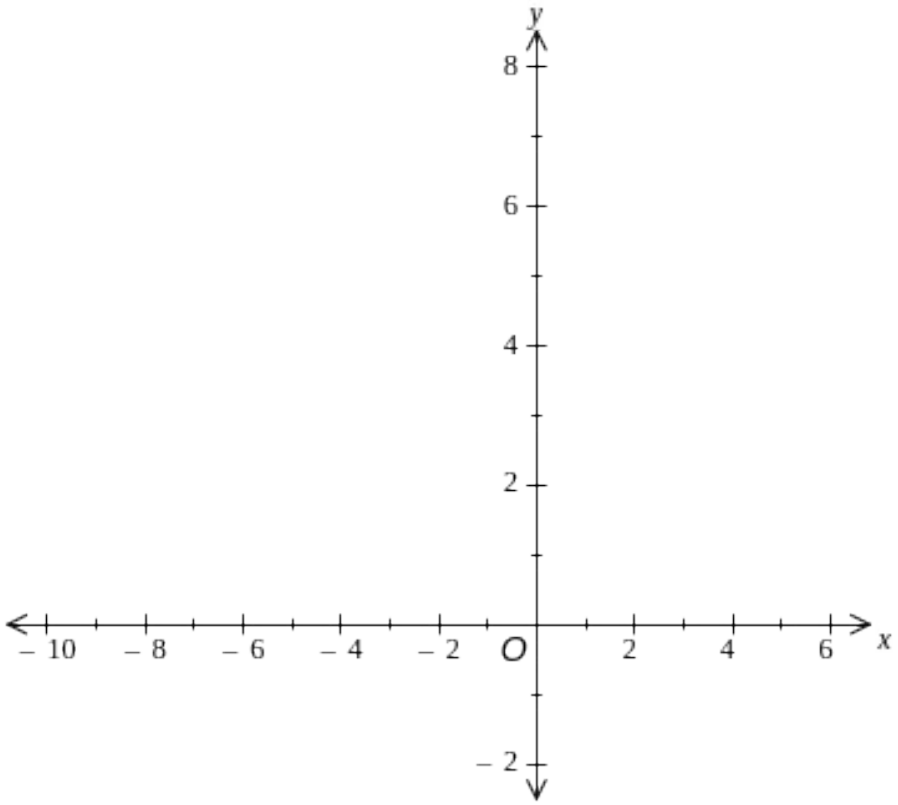
Question 1 (10 marks)

Consider the function f with rule $f(x) = \dfrac{x^2 - a|x| + b}{|x - 1|}$ where $a, b \in R$

a. State the equations of all potential asymptotes of f. 2 marks

b. Verify that f is not smooth at the point where $y = b$ except when a has one particular value, which you should state. 3 marks

c. Sketch the graph of $y = f(x)$ on the axes below when $a = 5$ and $b = 6$. Label asymptotes with their equations, and label axis intercepts with their coordinates. 3 marks



d. Consider the function g with rule $g(x) = \frac{x^2 - 5|x| + 6}{|x - 1|} - \arcsin(x - 1) - k$ for $k \in R$.

Find the values of k for which the graph of $y = g(x)$ has three x-intercepts. Give answers correct to four decimal places if an exact answer is not possible. 2 marks

Question 2 (12 marks)

Let $S = \{z : |z - 1| = 1, z \in C\}$.

a. Express S in cartesian form. 1 mark

b. Find the larger of the two areas bounded by S and $\{z : |z| \geq |z - 1|, z \in C\}$. 2 marks

c. All except one of the elements of S are elements of

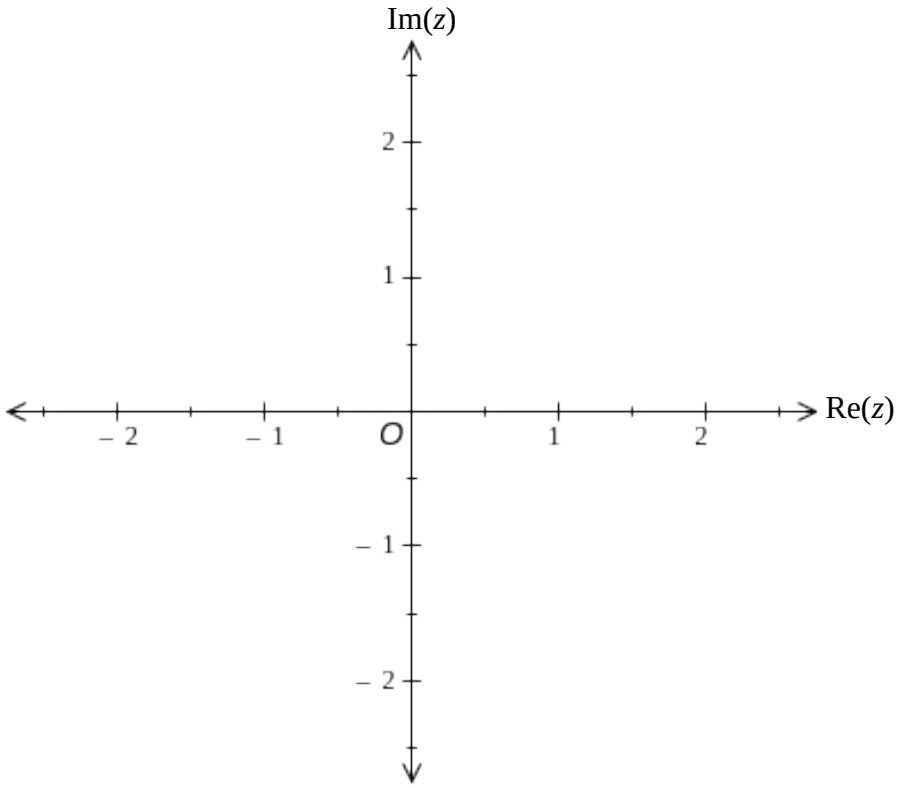
$\left\{z:\ a\text{Arg}\left(1-\frac{b}{z}\right)=\text{Arg}\left(z^2\right),\ z\in C\right\}$ where $a,b\in R^+$

Find the values of a and b. 2 marks

d. Prove that all except one of the elements of S are elements of $\left\{z:\ \text{Arg}(z-1)=\text{Arg}\left(z^2\right),\ z\in C\right\}$

2 marks

e. Sketch $\left\{z:\ \text{Arg}(z-1)=\text{Arg}\left(z^2\right),\ z\in C\right\}$ on the Argand diagram below. 3 marks



f. Let $z\in S\cap\left\{z:\ 3\text{Arg}\left(z^2\right)=2\text{Arg}\left(z^2-z\right),\ z\in C\right\}$. Find the values of $\text{Arg}(z)$. 2 marks

Question 3 (9 marks)

Consider the curve given by $y = \sqrt{3 - \tan^{-1}(2x)}$

a. The area bounded by the curve and the x-axis over the interval $x \in [0, 1]$ is rotated about the x-axis to form a solid of revolution.

Find, correct to four decimal places, the volume of this solid. 2 marks

b. Find the area of the surface formed when the part of the curve over the interval $x \in [-1, 0]$ is rotated about the y-axis. Give your answer correct to four decimal places. 3 marks

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c. Find the arclength of the curve over the interval $x \in [-1, 1]$. Give your answer correct to four decimal places. 2 marks

Consider the curves $y = f(x)$ and $y = g(x) = f(x) + k$ where $f(x) > 0$ over the interval $x \in [a, b]$ and $k \in R^+$

Let L be the arclength of $y = f(x)$ over $x \in [a, b]$

Let S_f and S_g be the respective areas of the surface formed when the part of each curve $y = f(x)$ and $y = g(x)$ over the interval $x \in [a, b]$ is rotated about the x-axis.

d. Express S_g in terms of S_f and L. 2 marks

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Question 4 (10 marks)

At the start of the year 2024 the *Gratuitous Fish Farm* releases 150 fish into a pond that originally contained no fish. Fish are removed from the pond at the end of each year.
The fish population, N, grows according to the model:

$$\frac{dN}{dt} = 2N\left(1 - \frac{N}{600}\right) - n$$

where t is the number of years after the release of the 150 fish and n is the number of fish removed at the end of each year.

a. Find the value(s) of n for which the rate at which the fish population initially grows is positive. 1 mark

b. Find the value(s) of n such that the solution to $\frac{dN}{dt} = 2N\left(1 - \frac{N}{600}\right) - n$ has the form:

$N = 5 + \frac{k}{t - c}$ where $c, k \in R$ 2 marks

c. Find, correct to the nearest integer, the value of n such that there are no fish in the pond after four years. 2 marks

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Suppose $n = 100$.

d. Without solving the differential equation:

i. Given that the graph of $N = N(t)$ has a point of inflection, find the N - coordinate of this point. 2 marks

ii. Briefly explain why N has a limiting value and find this value, correct to the nearest integer. 2 marks

e. Find, in years, the value of t when $N = 400$. Give your answer correct to two decimal places. 1 mark

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Question 5 (10 marks)

Let the vectors $\underset{\sim}{v} = -\underset{\sim}{i} - 2\underset{\sim}{j} + \underset{\sim}{k}$ and $\underset{\sim}{w} = 3\underset{\sim}{i} - 2\underset{\sim}{j} - \underset{\sim}{k}$.

a. Show that $\underset{\sim}{v}$ and $\underset{\sim}{w}$ are perpendicular to each other. 1 mark

A plane, Π, contains the point $(2, -2, -1)$ and the vectors $\underset{\sim}{v}$ and $\underset{\sim}{w}$ are parallel to Π.

b. Find a cartesian equation of Π. 2 marks

The position vector of an object is given by

$$\underset{\sim}{r} = 4\underset{\sim}{i} + 2\underset{\sim}{j} - 3\underset{\sim}{k} + \cos(2t)\underset{\sim}{v} + \sin(3t)\underset{\sim}{w}, \quad 0 \le t \le 2\pi$$

where time t is measured in minutes and distance is measured in decametres.

c. Show that Π contains the path followed by the object. 2 marks

d. Find the distance travelled by the object after one minute. Give your answer in decametres correct to four decimal places. 2 marks

The position vector of a second object is given by

$$\underset{\sim}{s} = \underset{\sim}{i} + \underset{\sim}{j} + 6\underset{\sim}{k} + t\left(-\underset{\sim}{i} + 6\underset{\sim}{j} - \underset{\sim}{k}\right), \quad t \geq 0$$

where time t is measured in minutes and distance is measured in decametres.

e. Find a cartesian equation of the path followed by the second object. 1 mark

f. Find, in decametres, the minimum distance of the second object from Π.

Give your answer in the form $\frac{a\sqrt{b}}{3}$ where $a, b \in N$ 2 marks

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Question 6 (9 marks)

Gilne Garden Supplies sells large and small bags of manure. It is known that the amount of manure in a small bag is normally distributed with a standard deviation of 0.75 kg.

a. The probability that the combined amount of manure in three small bags is more than 46 kg is approximately 0.6238.

Use an appropriate value of the standard normal random variable Z to find the mean amount of manure in a small bag. Give your answer in kg correct to two decimal places. 3 marks

It is known that the amount of manure in a large bag is normally distributed with a standard deviation of 1 kg. The mean amount of manure in a large bag is advertised to be 25 kg.

Stefan is the manure manager. He thinks that the true mean amount of manure in a large bag is not 25 kg and decides to apply a statistical test at the 3% level of significance.

b. Write down the null hypothesis, H_0, the alternative hypothesis, H_1, and the probability of a type I error for the test. 1 mark

Stefan takes a random sample of 32 large bags and finds that the total amount of manure is 807.4 kg.

c. Find the p value for the test correct to four decimal places. 2 marks

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Suppose that the true mean amount of manure in a large bag is 25.6 kg.

d. Calculate the probability, correct to three decimal places, of Stefan making a type II error in the statistical test. 3 marks

END OF EXAMINATION

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Multiple-Choice Answer Sheet

Student Name:

Shade the letter that corresponds to each correct answer.

Question	A	B	C	D
Question 1	[ ]	[ ]	[ ]	[ ]
Question 2	[ ]	[ ]	[ ]	[ ]
Question 3	[ ]	[ ]	[ ]	[ ]
Question 4	[ ]	[ ]	[ ]	[ ]
Question 5	[ ]	[ ]	[ ]	[ ]
Question 6	[ ]	[ ]	[ ]	[ ]
Question 7	[ ]	[ ]	[ ]	[ ]
Question 8	[ ]	[ ]	[ ]	[ ]
Question 9	[ ]	[ ]	[ ]	[ ]
Question 10	[ ]	[ ]	[ ]	[ ]
Question 11	[ ]	[ ]	[ ]	[ ]
Question 12	[ ]	[ ]	[ ]	[ ]
Question 13	[ ]	[ ]	[ ]	[ ]
Question 14	[ ]	[ ]	[ ]	[ ]
Question 15	[ ]	[ ]	[ ]	[ ]
Question 16	[ ]	[ ]	[ ]	[ ]
Question 17	[ ]	[ ]	[ ]	[ ]
Question 18	[ ]	[ ]	[ ]	[ ]
Question 19	[ ]	[ ]	[ ]	[ ]
Question 20	[ ]	[ ]	[ ]	[ ]

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VCE® Specialist Mathematics

Practice Written Examination 2

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Solution Pathway

Below are sample answers and solutions. Please consider the merit of alternative responses.

Specialist Mathematics Examination 2: Marking Scheme

SECTION A: Multiple-Choice Questions – Answers

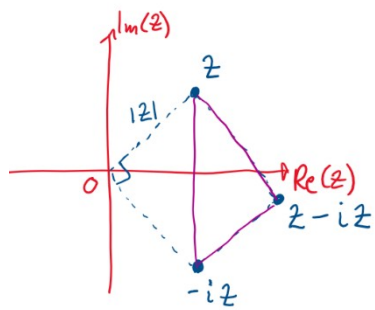
1.	C	**5.**	C	**9.**	A	**13.**	C	**17.**	C
2.	B	**6.**	D	**10.**	B	**14.**	D	**18.**	D
3.	A	**7.**	D	**11.**	C	**15.**	A	**19.**	C
4.	D	**8.**	B	**12.**	D	**16.**	B	**20.**	A

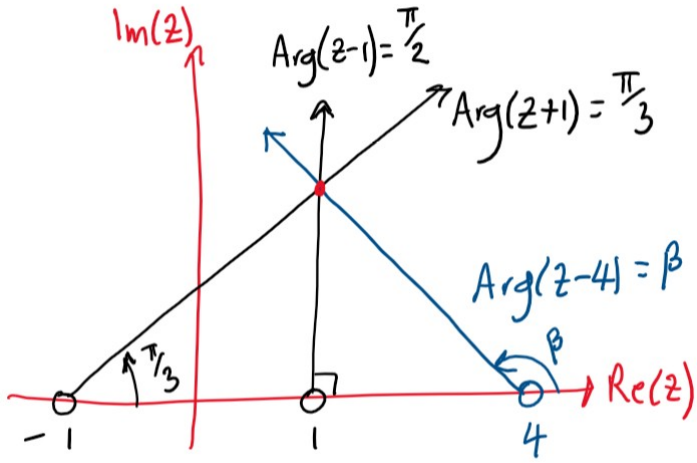
SECTION A: Multiple-Choice Questions – Solutions

Q 1	To prove that if $\underbrace{n^2 - 6n + 5 \text{ is even}}_{P}$ $\underbrace{\text{then}}_{\Rightarrow}$ $\underbrace{n \text{ is odd}}_{Q}$, it is assumed for the sake of contradiction that $\underbrace{n^2 - 6n + 5 \text{ is even}}_{P}$ $\underbrace{\text{and}}_{\wedge}$ $\underbrace{n \text{ is even}}_{\neg Q}$	**C**

Q 2	• $x=6$ is a vertical asymptote therefore $x=6$ is a root of $ax^2+b\|x\|+c$: $36a+6b+c=0$ …. (1) • Let $g(x)=\dfrac{x^3-1}{ax^2+bx+c}$ and $f(x)=\dfrac{\|x\|^3-1}{ax^2+b\|x\|+c}=g(\|x\|)$ From a CAS: $g(x)=\dfrac{1}{a}x-\dfrac{b}{a^2}+\dfrac{\text{Linear}}{ax^2+bx+c}$ Therefore $y=g(x)$ has an oblique asymptote $y=\dfrac{1}{a}x-\dfrac{b}{a^2}$ • **Case 1:** $x\geq 0$: $f(x)=g(x)$ therefore $y=\dfrac{1}{a}x-\dfrac{b}{a^2}$ is a diagonal asymptote of $y=f(x)$. Compare $y=\dfrac{1}{a}x-\dfrac{b}{a^2}$ with $y=-2x+14$: $\dfrac{1}{a}=-2\Rightarrow a=-\dfrac{1}{2}$ $-\dfrac{b}{a^2}=14\Rightarrow b=-14a^2=-\dfrac{7}{2}$. Substitute $a=-\dfrac{1}{2}$ and $b=-\dfrac{7}{2}$ into (1): $c=39$. No corresponding option. • **Case 2:** $x<0$: $f(x)=g(-x)$ therefore $y=-\dfrac{1}{a}x-\dfrac{b}{a^2}$ is a diagonal asymptote of $y=f(x)$ Compare $y=-\dfrac{1}{a}x-\dfrac{b}{a^2}$ with $y=-2x+14$: $-\dfrac{1}{a}=-2\Rightarrow a=\dfrac{1}{2}$ $-\dfrac{b}{a^2}=14\Rightarrow b=-14a^2=-\dfrac{7}{2}$. Substitute $a=\dfrac{1}{2}$ and $b=-\dfrac{7}{2}$ into (1): $c=3$. **Option B**.	**B**

	Discussion of Case 2: It follows that $g(x)=\dfrac{x^3-1}{\frac{1}{2}x^2-\frac{7}{2}x+3}=\dfrac{2(x-1)\left(x^2+x+1\right)}{(x-1)(x-6)}=\dfrac{2\left(x^2+x+1\right)}{x-6}$, $x\neq 1$ Therefore $y=g(x)$ has a 'hole' at $x=1$.	

Q 3	Horizontal asymptotes: $y = a - b^2\left(\frac{\pi}{2}\right)$ and $y = a - b^2\left(-\frac{\pi}{2}\right) = a + b^2\left(\frac{\pi}{2}\right)$. It is therefore required that either: **Case 1:** $a - b^2\left(\frac{\pi}{2}\right) \geq 0$ and $a > 0$ therefore $b^2 \leq \frac{2a}{\pi}$ and $a > 0$ or **Case 2:** $a + b^2\left(\frac{\pi}{2}\right) \leq 0$ and $a < 0$ therefore $b^2 \leq -\frac{2a}{\pi}$ and $a < 0$ Both cases are equivalent to $b^2 \leq \left\lvert\frac{2a}{\pi}\right\rvert$	**A**
Q 4	The geometric interpretation of $-iz$ is the rotation of z by 90^0 in a clockwise direction. Therefore the area of the triangle is half the area of a square whose vertices are at 0, z, $-iz$, $z - iz$:  By inspection the length of a side of the square is $\lvert z \rvert$ therefore the area of the square is $\lvert z \rvert^2$ therefore the area of the triangle is $\frac{1}{2}\lvert z \rvert^2$.	**D**

Q 5	Let $z = x + iy$, $x, y \in R$. The rays $\text{Arg}(z+1) = \frac{\pi}{3}$, $\text{Arg}(z-1) = \frac{\pi}{2}$ and $\text{Arg}(z-4) = \beta$ intersect at the point where $x = 1$:  The cartesian equation of $\text{Arg}(z+1) = \frac{\pi}{3}$ is $y = \sqrt{3}x + \sqrt{3}$, $x > -1$ Substitute $x = 1$: $y = 2\sqrt{3}$ Therefore the rays $\text{Arg}(z+1) = \frac{\pi}{3}$, $\text{Arg}(z-1) = \frac{\pi}{2}$ and $\text{Arg}(z-4) = \beta$ intersect at $z = 1 + 2\sqrt{3}i$ Substitute $z = 1 + 2\sqrt{3}i$ into $\text{Arg}(z-4) = \beta$: $\beta = \text{Arg}\left(-3 + 2\sqrt{3}i\right) = \pi - \arctan\left(\frac{2\sqrt{3}}{3}\right)$	**C**

Q 6 **D**

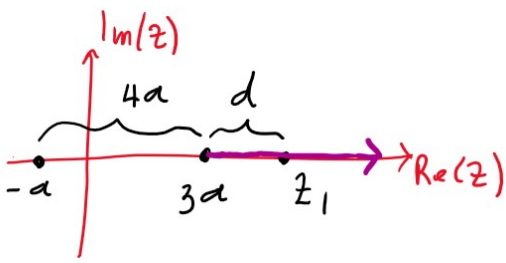
a is real therefore $3a$ and $-a$ lie on the real axis.

Geometric interpretation:

$|z+a|$ is the distance of z from $-a$, $|z-3a|$ is the distance of z from $3a$

The distance between $-a$ and $3a$ is $4a$

Let $z=z_1 \in R$ where $z_1 > 3a$. Let $|z_1-3a|=d \Rightarrow |z_1+a|=4a+d$:



Then $|z_1+a|-|z_1-3a|=(4a+d)-d=4a$

Therefore $|z+a|-|z-3a|=4a$ defines a ray.

Discussion: By inspection, the equation of the ray is $y=0$, $x \geq 3a$. The terminus of the ray has coordinates $(3a,\ 0)$ and is included as part of the ray.

Algebraic calculation:

Substitute $z=x+iy$, $x, y \in R$, into $|z+a|-|z-3a|=k$:

$$\sqrt{(x+a)^2+y^2}-\sqrt{(x-3a)^2+y^2}=k$$

$$\Rightarrow \sqrt{(x+a)^2+y^2}=k+\sqrt{(x-3a)^2+y^2}$$

$$\Rightarrow (x+a)^2+y^2=k^2+(x-3a)^2+y^2+2k\sqrt{(x-3a)^2+y^2}$$

$$\Rightarrow 8ax-8a^2-k^2=2k\sqrt{(x-3a)^2+y^2}$$

Note the implied restriction:

RHS ≥ 0 therefore LHS $=8ax-8a^2-k^2 \geq 0 \Rightarrow x \geq \dfrac{8a^2+k^2}{8a}$

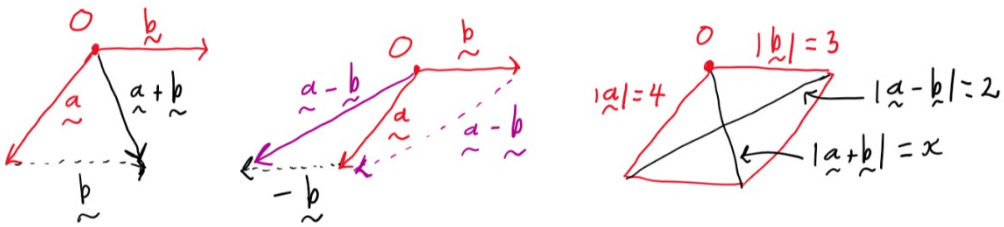
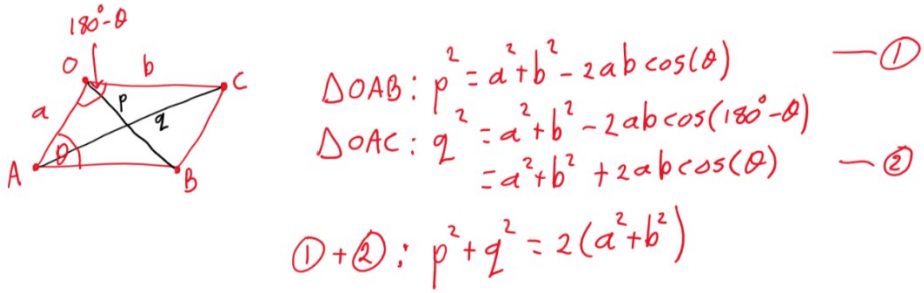
	$\Rightarrow \left(8ax - 8a^2 - k^2\right)^2 = 4k^2\left((x-3a)^2 + y^2\right)$ $\Rightarrow (4a-k)(4a+k)(2x+k-2a)(2x-k-2a) = 4k^2y^2$, $x \geq \frac{8a^2+k^2}{8a}$ If $k = 4a$ then a ray is obtained: $0 = 4k^2y^2 \Rightarrow y = 0$, $x \geq 3a$ If $k > 4a$: $x > 3a$ and there is no solution since LHS $= (-ve)(+ve)(+ve)(+ve) < 0$ and RHS > 0	
Q 7	The pseudocode applies Euler's method to find an approximate solution to the differential equation $\frac{dx}{dt} = \frac{t^2+1}{x^3+1} = f(t,x)$ given that $t_0 = a$ and $x_0 = 3$ The number of iterations is $n = 10$ therefore $t_{10} = b = 4$, the value of the step size is $\frac{b-a}{n} = \frac{4-a}{10}$ and $x_{10} = 3.504$ (output correct to three decimal places). Substitute the value of a in each option into the above data and execute Euler's Method on a CAS.	**D**

Q 8	$\frac{dm}{dt} =$ (rate at which salt enters the tank) - (rate at which salt leaves the tank). • Rate at which salt enters the tank = (rate of inflow) (concentration of salt in inflow) = (5 litres per minute) (6 kg per litre) = 30 kg per minute. • Rate at which salt leaves the tank = (rate of outflow) (concentration of salt in outflow) = (7 litres per minute) (concentration of salt in outflow). • After t minutes have elapsed: Concentration of salt in outflow $=\frac{m}{V}$ where V is the volume of solution in the tank after t minutes: $V =$ (initial volume) + (volume added after t minutes via inflow) - (volume removed after t minutes via outflow) = 90 litres + 5t litres - 7t litres $=90 - 2t$ litres. Therefore: concentration of salt in outflow $=\frac{m}{V} = \frac{m}{90 - 2t}$ kg per litre. Therefore rate at which salt leaves the tank = (7 litres per minute) (concentration of salt in outflow) $=(7)\left(\frac{m}{90 - 2t}\right) \quad = \frac{7m}{90 - 2t}$ kg per minute $\frac{dm}{dt} =$ (rate at which salt enters the tank) - (rate at which salt leaves the tank) $=(30) - \left(\frac{7m}{90 - 2t}\right) = 30 - \frac{7m}{90 - 2t}$	**B**

	• Differential equation for the concentration *c* of salt in the tank at time *t*: $c=\frac{m}{V}=\frac{m}{90-2t} \quad \Rightarrow m=c(90-2t)$ $\Rightarrow \frac{dm}{dt}=\frac{d}{dt}(c(90-2t)) \quad =\frac{dc}{dt}(90-2t)+c(-2) \quad =(90-2t)\frac{dc}{dt}-2c$ Product Rule Substitute $\frac{dm}{dt}=30-\frac{7m}{90-2t}$: $(90-2t)\frac{dc}{dt}-2c=30-\frac{7c(90-2t)}{90-2t} \quad \Rightarrow (90-2t)\frac{dc}{dt}=30-5c$	
Q 9	The value of $\frac{dy}{dx}$ at $t=2$ is required. When $t>\sqrt{3}$: $x(t)=\frac{\lvert 3-t^2\rvert}{t+1}=\frac{t^2-3}{t+1}$, $y(t)=-\frac{4}{t+2}$. Use a CAS to get the values of $\frac{dx}{dt}$ and $\frac{dy}{dt}$ when $t=2$: $\frac{dy}{dx}=\frac{dy}{dt}\times\frac{dt}{dx}=\frac{9}{44}$ when $t=2$	**A**

Q 10	Consider $J_n = \int \sec^n(x)\,dx$ and use integration by parts: $\int u\frac{dv}{dx}dx = uv - \int v\frac{du}{dx}dx$ Let $u = \sec^{n-2}(x)$ $\Rightarrow \frac{du}{dx} = (n-2)\sec(x)^{n-3}\sin(x)\sec^2(x) = (n-2)\sec(x)^{n-1}\sin(x)$ Let $\frac{dv}{dx} = \sec^2(x) \Rightarrow v = \tan(x)$ $J_n = \underbrace{\sec^{n-2}(x)}_{u}\underbrace{\tan(x)}_{v} - \int \underbrace{\tan(x)}_{v}\underbrace{(n-2)\sec^{n-1}(x)\sin(x)}_{\frac{du}{dx}}\,dx$ $= \sec^{n-2}(x)\tan(x) - (n-2)\int \sin^2(x)\sec^n(x)\,dx$ $= \sec^{n-2}(x)\tan(x) - (n-2)\int \left(1-\cos^2(x)\right)\sec^n(x)\,dx$ $= \sec^{n-2}(x)\tan(x) - (n-2)\left[\int \sec^n(x)\,dx - \int \sec^{n-2}(x)\,dx\right]$ $= \sec^{n-2}(x)\tan(x) - (n-2)\left[J_n - J_{n-2}\right]$ $= \sec^{n-2}(x)\tan(x) - (n-2)J_n + (n-2)J_{n-2}$ $\Rightarrow J_n = \frac{1}{n-1}\sec^{n-2}(x)\tan(x) + \frac{n-2}{n-1}J_{n-2}$ $(n \neq 1)$ $\Rightarrow I_n = \frac{1}{n-1}\left[\sec^{n-2}(x)\tan(x)\right]_0^{\frac{\pi}{3}} + \frac{n-2}{n-1}I_{n-2} = \frac{2^{n-2}}{n-1}\sqrt{3} + \frac{n-2}{n-1}I_{n-2}$	**B**

Q 11	The area of the surface generated by rotating the curve with equation $y = \tan^{-1}(x)$ about the line $x = 1$ from the points where $x = 2$ to $x = 4$ is equal to the surface generated by rotating the curve with equation $y = \tan^{-1}(x+1)$ about the *y*-axis from the points where $x = 1$ to $x = 3$. Area of the surface generated by rotating the curve $y = f(x)$ about the *y*-axis: $S = 2\pi \int_{y=a}^{y=b} x\sqrt{1+\left(\frac{dx}{dy}\right)^2}\, dy$ $y = \tan^{-1}(x+1) \Rightarrow \tan(y) = x+1 \Rightarrow x = \tan(y) - 1$ $x = 1 \Rightarrow y = \tan^{-1}(2)$ $\quad$ $x = 3 \Rightarrow y = \tan^{-1}(4)$ $S = 2\pi \int_{\tan^{-1}(2)}^{\tan^{-1}(4)} (\tan(y) - 1)\sqrt{1+\sec^4(y)}\, dy$	**C**

Q 12	The vectors $\underset{\sim}{a}$ and $\underset{\sim}{b}$ form a parallelogram with pairs of opposite sides of length $\lvert\underset{\sim}{a}\rvert=4$ and $\lvert\underset{\sim}{b}\rvert=3$ and diagonals of length $\lvert\underset{\sim}{a}-\underset{\sim}{b}\rvert=2$ and $\lvert\underset{\sim}{a}+\underset{\sim}{b}\rvert=x$:  For the parallelogram with pairs of opposite sides of length a and b and diagonals of length p and q: $p^2+q^2=2(a^2+b^2)$.  $p^2+q^2=2(a^2+b^2) \Rightarrow 2^2+x^2=2(4^2+3^2) \Rightarrow x=\lvert\underset{\sim}{a}+\underset{\sim}{b}\rvert=\sqrt{46}$	**D**
Q 13	Let the angle between $\underset{\sim}{a}$ and $\underset{\sim}{b}$ be θ: • $\underset{\sim}{a}\cdot\underset{\sim}{b}=4 \Rightarrow \lvert\underset{\sim}{a}\rvert\lvert\underset{\sim}{b}\rvert\cos(\theta)=4$. …. (1) • $\lvert\underset{\sim}{a}\times\underset{\sim}{b}\rvert=7 \Rightarrow \lvert\underset{\sim}{a}\rvert\lvert\underset{\sim}{b}\rvert\sin(\theta)=7$. …. (2) • $\frac{(2)}{(1)}$: $\tan(\theta)=\frac{7}{4} \Rightarrow \theta\approx 60.3^0$	**C**
Q 14	$v=\frac{dx}{dt}=\tan^{-1}(x) \Rightarrow \frac{dt}{dx}=\frac{1}{\tan^{-1}(x)}$ and $x=2$ when $t=3$.	**D**

	The integral solution must be used: $t = \int_2^x \frac{1}{\tan^{-1}(w)}\,dw + 3$. Substitute $x = 5$: $t = \int_2^5 \frac{1}{\tan^{-1}(w)}\,dw + 3 \approx 5.359$	
Q 15	The acceleration of the object is constant therefore the straight line motion formulae for constant acceleration can be used: • Motion after 10 seconds: Data: $u = 6\ \text{ms}^{-1}$, $v = -8\ \text{ms}^{-1}$, $t = 10$ seconds, $a = ?$ $v = u + at \quad \Rightarrow -8 = 6 + a(10) \quad \Rightarrow a = -1.4\ \text{ms}^{-2}$ • Motion after 12 seconds: Data: $u = 6\ \text{ms}^{-1}$, $t = 12$ seconds, $a = -1.4\ \text{ms}^{-2}$, $x = ?$ $x = ut + \frac{1}{2}at^2 \quad = 6(12) + \frac{1}{2}(-1.4)(12)^2 \quad = -28.8$ m Therefore the distance is 28.8 metres.	**A**

Q 16 **B**

- Time at which the aircraft drops the package: $540 = 600 - 3t \Rightarrow t = 20$ seconds.
- Initial conditions for the motion of the package:

$\underset{\sim}{r}_p(0) = \underset{\sim}{r}(20) = 1000\,\underset{\sim}{i} + 540\,\underset{\sim}{j}$, $\dot{\underset{\sim}{r}}_p(0) = \dot{\underset{\sim}{r}}(20) = 50\,\underset{\sim}{i} - 3\,\underset{\sim}{j}$

Method 1:

Solve $\ddot{\underset{\sim}{r}}_p(t) = -9.8\,\underset{\sim}{j}$ (since the air resistance acting on the package is negligible) subject to the above initial conditions (either using a CAS or 'by hand'):

$$\underset{\sim}{r}_p(t) = (1000 + 50t)\,\underset{\sim}{i} + \left(540 - 3t - 4.9t^2\right)\underset{\sim}{j}$$

Therefore $x_p(t) = 1000 + 50t$ and $y_p(t) = 540 - 3t - 4.9t^2$

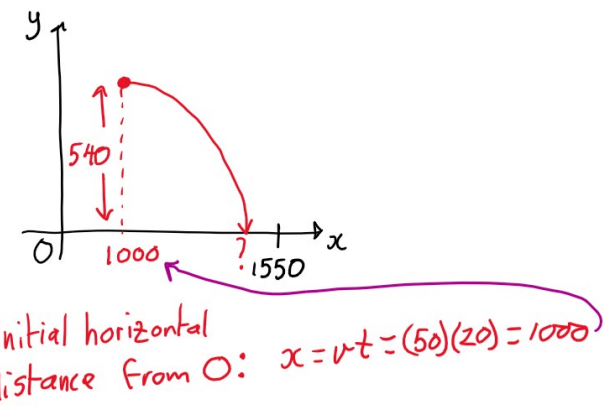
- Time at which the package lands:

$y_p(t) = 540 - 3t - 4.9t^2 = 0$ (and $t > 0$) $\Rightarrow t = 10.1962$

- Horizontal distance of package from O:

$x_p(10.1962) = 1000 + 50(10.1962) = 1509.81$ m

Method 2:



- Acceleration of the package is constant therefore the straight line motion formulae for constant acceleration can be used:

Horizontal motion ($\rightarrow$ +ve):
$u = 50$ ms^{-1}, $a = 0$ ms^{-2}
$y = 540$ m
$t = ?$ 10.1962, $x = ?$

Vertical motion ($\downarrow$ +ve):
$u = 3$ ms^{-1}, $a = 9.8$ ms^{-2},
$t = ?$

	$x = ut + \frac{1}{2}at^2 = (50)(10.1962) = 509.81$ $\quad y = ut + \frac{1}{2}at^2 \Rightarrow 540 = 3t + 4.9t^2$ $\Rightarrow t \approx 10.1962$ • Therefore horizontal distance of package from O $= 1000 + 509.81 = 1509.81$ • Therefore distance short of the drop position $= 1550 - 1509.81 = 40.19$ m	
Q 17	• Determine the value(s) of k for which the planes $2x + (k-4)y + (3-k)z = 1$, $2y + (k-3)z = 2$, $x - 2y + z = 1$ do not intersect at a unique point: $\det\begin{vmatrix} 2 & k-4 & 3-k \\ 0 & 2 & k-3 \\ 1 & -2 & 1 \end{vmatrix} = 0 \quad \Rightarrow k = -1, 2$ • Test each solution. $k = -1$: $2x - 5y + 4z = 1$, $y - 2z = 1$, $x - 2y + z = 1$ Solve simultaneously (use a CAS): Infinite number of solutions which corresponds to a line. $k = 2$: $2x - 2y + z = 1$, $2y - z = 2$, $x - 2y + z = 1$ Solve simultaneously (use a CAS): No solution.	**C**

Q 18	• $\frac{x-1}{3}=\frac{3-y}{\alpha}=\frac{z+1}{4} \Rightarrow x=3t+1,\ y=-\alpha t+3,\ z=4t-1,\ t\in R$ Therefore a vector in the direction of the line is $3\underset{\sim}{i}-\alpha\underset{\sim}{j}+4\underset{\sim}{k}$ • $\frac{x+1}{\beta}=\frac{y-2}{3}=\frac{z-1}{5} \Rightarrow x=\beta s-1,\ y=3s+2,\ z=5s+1,\ s\in R$ Therefore a vector in the direction of the line is $\beta\underset{\sim}{i}+3\underset{\sim}{j}+5\underset{\sim}{k}$ • Therefore a vector normal to each line is (use a CAS) is $\left(3\underset{\sim}{i}-\alpha\underset{\sim}{j}+4\underset{\sim}{k}\right)\times\left(\beta\underset{\sim}{i}+3\underset{\sim}{j}+5\underset{\sim}{k}\right)=(-12-5\alpha)\underset{\sim}{i}+(4\beta-15)\underset{\sim}{j}+(9+\alpha\beta)\underset{\sim}{k}$ • Therefore $3\underset{\sim}{i}-7\underset{\sim}{j}+\gamma\underset{\sim}{k}=(-12-5\alpha)\underset{\sim}{i}+(4\beta-15)\underset{\sim}{j}+(9+\alpha\beta)\underset{\sim}{k}$ Equate $\underset{\sim}{i}$-components: $3=-12-5\alpha \Rightarrow \alpha=-3$ Equate $\underset{\sim}{j}$-components: $-7=4\beta-15 \Rightarrow \beta=2$ Equate $\underset{\sim}{k}$-components: $\gamma=9+(-3)(2)=3$	**D**
Q 19	• $\text{sd}\left(\overline{X}\right)=\frac{\text{sd}(X)}{\sqrt{n}}$. • $\text{Var}(X)=E\left(X^2\right)-\left(E(X)\right)^2=\int_0^1 x^2(2x)\,dx-\left(\int_0^1 x(2x)\,dx\right)^2=\frac{1}{18}$ $\Rightarrow \text{sd}(X)=\frac{1}{\sqrt{18}}$ • $\text{sd}\left(\overline{X}\right)=\frac{\text{sd}(X)}{\sqrt{n}}=\frac{\frac{1}{\sqrt{18}}}{\sqrt{80}}=\frac{\sqrt{10}}{120}$	**C**

Q 20	**Method 1:** Use a CAS to check the confidence interval arising from the value of n in each option: $n = 92$. **Method 2:** $14572 = \bar{x} + k\frac{\sigma}{\sqrt{n}}$ …. (1) $14428 = \bar{x} - k\frac{\sigma}{\sqrt{n}}$ …. (2) $(1) + (2) \Rightarrow \bar{x} = 14500$ For a 90% confidence interval: $\Pr(-k < Z < k) = 0.95 \Rightarrow k \approx 1.64485$ (use a CAS). Substitute $\bar{x} = 14500$, $k = 1.64485$ and $\sigma = 420$ into either equation (1) or equation (2) and solve for n (use a CAS and round to the nearest integer): $n = 92$.	**A**

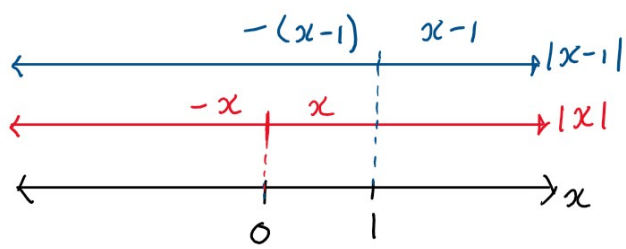
SECTION B: Solutions

Question 1

Part a.:

Note: These are only potential asymptotes because the asymptotic behaviour of f depends on the values of a and b. Some of these asymptotes do not exist for particular values of a and b (for example, f has no asymptotes when $a = 0$ and $b = -1$).

- Potential vertical asymptote: $x = 1$
- Asymptotic behaviour as $x \to \pm\infty$:



To consider $x \to +\infty$ it is therefore necessary to get the rule for $y = f(x)$ when $x \geq 1$:

$$y = f(x) = \frac{x^2 - ax + b}{x - 1} = x + 1 - a + \frac{b - a + 1}{x - 1}$$

Use a CAS or 'by hand'

$x \to +\infty$: $y \sim x + 1 - a$

To consider $x \to -\infty$ it is therefore necessary to get the rule for $y = f(x)$ when $x < 0$:

$$y = f(x) = \frac{x^2 - a(-x) + b}{-(x-1)} = \frac{x^2 + ax + b}{-x + 1} = -x - 1 - a - \frac{a + b + 1}{x - 1}$$

Use a CAS or 'by hand'

$x \to -\infty$: $y \sim -x - 1 - a$

Three correct potential asymptotes: $x = 1$, $y = x + 1 - a$, $y = -x - 1 - a$ Do **not** accept $y = \lvert x + 1 \rvert - a$ in place of $y = x + 1 - a$ and $y = -x - 1 - a$	**2 marks**

(because $y = x + 1 - a$ is sometimes a diagonal asymptote when $y = -x - 1 - a$ is not, and vice versa).	
Two correct potential asymptotes.	**1 mark**
One or no correct potential asymptotes.	**0 marks**

Part b.:

$$y = b \Rightarrow \frac{x^2 - a|x| + b}{|x - 1|} = b \Rightarrow x = 0$$

For f to be 'smooth' at $x = 0$ it is necessary and sufficient that $f'(x)$ is continuous at $x = 0$

It is therefore necessary and sufficient that:

1. $\lim_{x \to 0} f'(x)$ exists. 2. $\lim_{x \to 0} f'(x) = f'(0)$

1. Existence of $\lim_{x \to 0} f'(x)$

From a CAS:

$$\lim_{x \to 0^+} f'(x) = b - a \qquad \lim_{x \to 0^-} f'(x) = a + b$$

$$\lim_{x \to 0^+} f'(x) = \lim_{x \to 0^-} f'(x) \quad \Rightarrow b - a = a + b \quad \Rightarrow a = 0 \quad b \in R$$

and

$\lim_{x \to 0^+} f'(x) = b - a$ $\lim_{x \to 0^-} f'(x) = a + b$ $a = 0$ and $b \in R$	**1 mark**

2. $a=0$ and $b \in R \Rightarrow \lim_{x \to 0} f'(x) = b$. It must now be checked that $f'(0) = b$

$a=0$ and $x \to 0 \Rightarrow x < 1$ therefore $f(x) = \frac{x^2+b}{-(x-1)} = \frac{x^2+b}{-x+1}$

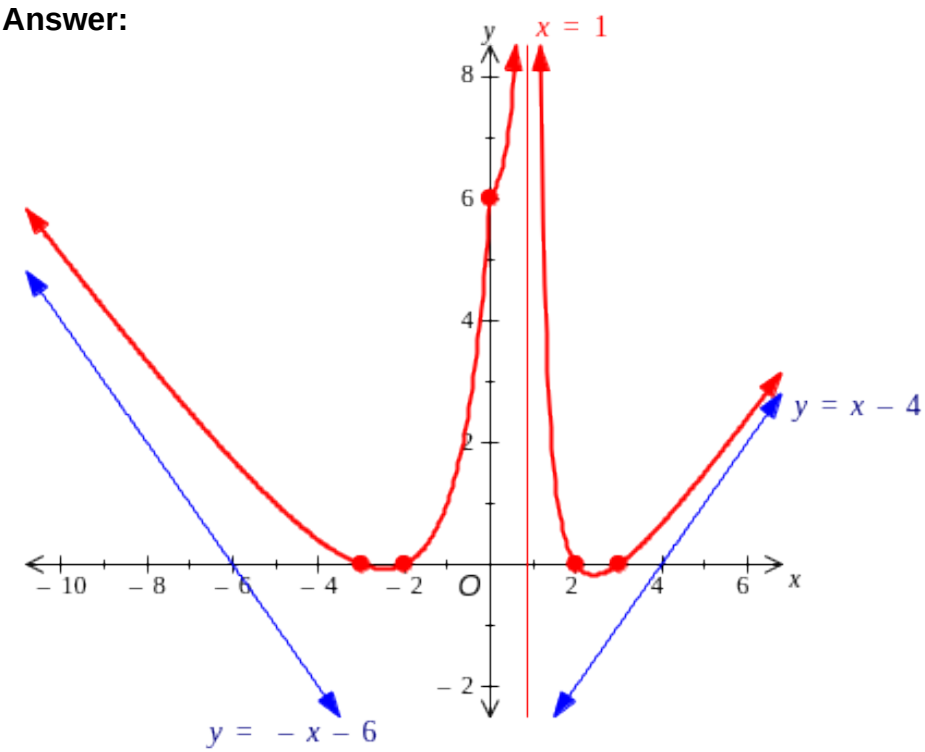
$f'(0) = b$

From a CAS:

$\lim_{x \to 0} f'(x) = b$ $\quad f(x) = \frac{x^2+b}{-x+1}$ $\quad f'(0) = b$ Therefore not 'smooth' at the point where $y = b$ unless $a = 0$	**1 mark**

Part c.:

Answer:



Shape: There must be a 'corner' at the *y*-intercept.	**1 mark**

The turning point in the interval $(2,\ 3)$ must be lower than the turning point in the interval $(-3,\ -2)$.	
Asymptotes: $x=1$, $y=x-4$, $y=-x-6$.	**1 mark**
Axis intercepts: $(2,\ 0)$, $(3,\ 0)$, $(-2,\ 0)$, $(-3,\ 0)$, $(0,\ 6)$.	**1 mark**

Calculations:

- Asymptotes are found from **Part a.**

Vertical asymptote: $x=1$

Diagonal asymptotes: Substitute $a=5$ into $y=x+1-a$ and $y=-x-1-a$

- Axis intercepts:

y-intercept: Substitute $x=0$ x-intercepts: Use a CAS to solve $\dfrac{x^2-5|x|+6}{|x-1|}=0$

- Shape: From **Part b.** the graph is not smooth at $x=0$ (that is, the y-intercept) because $a\neq 0$. Therefore there is a 'corner' at $x=0$

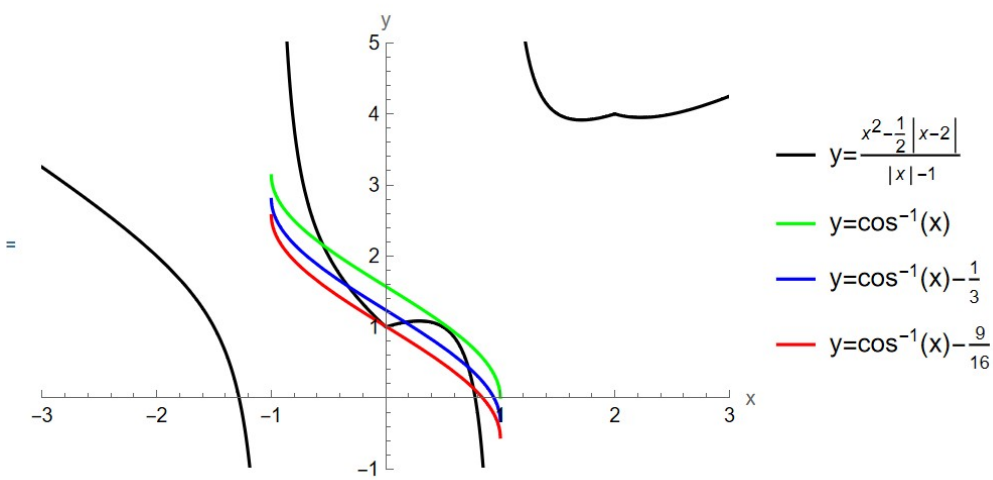
Part d.:

- $$\frac{x^2 - \frac{1}{2}|x-2|}{|x|-1} - \arccos(x) - k = 0 \quad \Rightarrow \frac{x^2 - \frac{1}{2}|x-2|}{|x|-1} = \arccos(x) + k$$

- Consider the graphs of $y = f(x) = \dfrac{x^2 - \frac{1}{2}|x-2|}{|x|-1}$ and $y = h(x) = \arccos(x) + k$

It is required to find the values of k for which these graphs have three intersection points

The value of k controls the vertical translation of $y = \arccos(x-1) + k$:

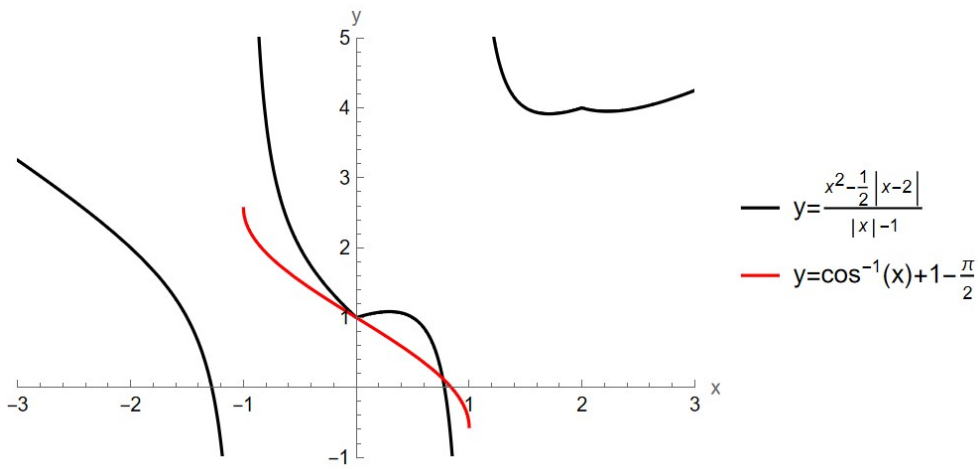


- Minimum value of k:

By inspection the value of k needs to be greater than the value of k such that point of inflection of $h(x) = \arccos(x) + k$ coincides with the 'corner' of $f(x) = \dfrac{x^2 - \frac{1}{2}|x-2|}{|x|-1}$ at (1, 0).

It is therefore required that the y-coordinate of the point of inflection of $y = \arccos(x) + k$ is equal to 1:

$h(0) = 1$: $\frac{\pi}{2} + k = 1 \quad \Rightarrow k = 1 - \frac{\pi}{2}$



- Maximum value of k:

By inspection there is some value $k = \alpha$ such that $f(x) = \dfrac{x^2 - \frac{1}{2}|x-2|}{|x|-1}$ and $h(x) = \arccos(x) + k$ have a common tangent in the interval $0 < x < 1$

There are therefore three intersection points when $1 - \dfrac{\pi}{2} < k \le \alpha$

$f(x) = \dfrac{2x^2 + x - 2}{2(x-1)}$ for $0 < x < 1$

Let $x = \beta$ at the point of common tangency. It is required that

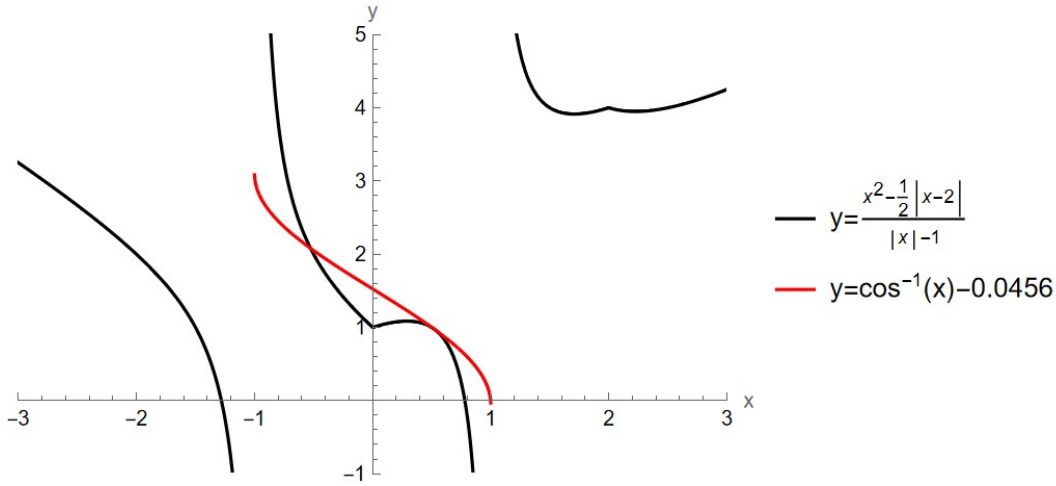
$f(\beta) = h(\beta)$ …. (1)

$f'(\beta) = h'(\beta)$ …. (2)

and $0 < \beta < 1$

Use a CAS to solve equations (1) and (2) simultaneously for α: $\alpha \approx -0.045612$

Note: $\beta \approx 0.520072$



$1-\frac{\pi}{2}<k\leq-0.0456$. **Note:** The correct maximum value of k, rounded to four decimal places, is -0.0457 and should be accepted (rounding UP is required). This is because $y=\arccos(x)-0.0456$ is above $y=\frac{x^2-\frac{1}{2}\lvert x-2\rvert}{\lvert x\rvert-1}$ on the interval $(0,\ 1)$	**2 marks:** 1 mark for the minimum value of k. 1 mark for the maximum value of k.

Question 2

Part a.:

$|z-1|=1$ is a circle with radius $r=1$ and centre at $z=1$:

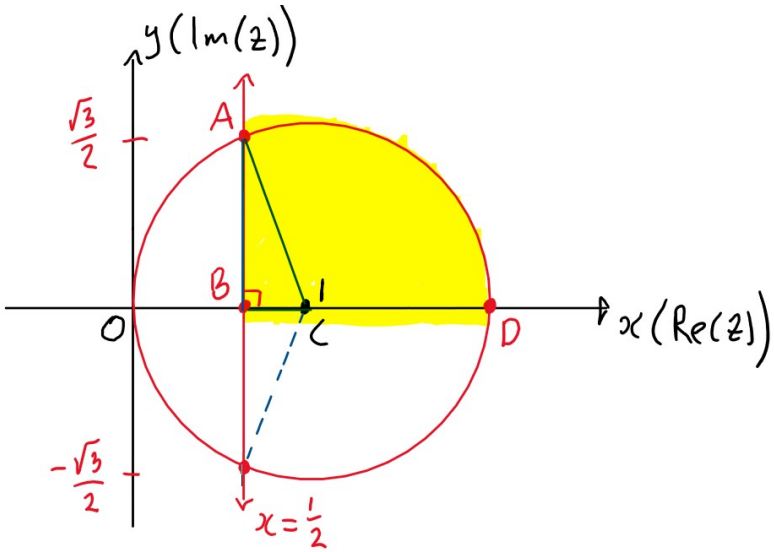
Answer: $(x-1)^2+y^2=1$	**1 mark**

Part b.:

- $|z|=|z-1|$ is the perpendicular bisector of the line segment joining $z=0$ and $z=1$: $x=\frac{1}{2}$
- Solve $(x-1)^2+y^2=1$ and $x=\frac{1}{2}$ simultaneously to get coordinates of intersection points:

Answer: $\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$ and $\left(\frac{1}{2}, \frac{-\sqrt{3}}{2}\right)$	**1 mark**

- Required area $=2(\text{Area of sector } ACD+\text{Area of triangle } ABC)$:


y (Im(z))
x (Re(z))
A
B
C
D
O
1
$\frac{\sqrt{3}}{2}$
$-\frac{\sqrt{3}}{2}$
$x = \frac{1}{2}$

- Area of sector ACD :

$$\cos(\angle ACB) = \frac{BC}{AC} = \frac{1}{2} \quad \Rightarrow \angle ACB = \frac{\pi}{3} \quad \Rightarrow \angle DCB = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$$

$$ACD = \frac{1}{3}(\text{Area of circle}) = \frac{\pi}{3}$$

Area of sector

Area of sector $ACD = \frac{\pi}{3}$	**1 mark**

- Area of triangle

$$ABC = \frac{1}{2}\left(\frac{1}{2}\right)\left(\frac{\sqrt{3}}{2}\right) = \frac{\sqrt{3}}{8}$$

$$= 2\left(\frac{\pi}{3} + \frac{\sqrt{3}}{8}\right)$$

- Required area

Answer: $\frac{2\pi}{3} + \frac{\sqrt{3}}{4}$ Accept $\frac{8\pi + 3\sqrt{3}}{12}$	**1 mark**

Part c.:

- $z=0$ is the exceptional element of $S=\{z:\ |z-1|=1,\ z\in C\}$ that is not an element of

$$\left\{z:\ a\text{Arg}\left(1-\frac{b}{z}\right)=\text{Arg}\left(z^2\right),\ z\in C\right\}$$

- All elements $S\setminus\{0\}$ are elements of $\left\{z:\ a\text{Arg}\left(1-\frac{b}{z}\right)=\text{Arg}\left(z^2\right),\ z\in C\right\}$

Therefore substitute two convenient elements of $S\setminus\{0\}$ (use the answer to **Part a.**). For example:

$z=1+i$: $a\text{Arg}\left(1-\frac{b}{1+i}\right)=\text{Arg}\left((1+i)^2\right)$ …. (1)

$z=\frac{1}{2}+\frac{\sqrt{3}}{2}i$: $a\text{Arg}\left(1-\frac{b}{\frac{1}{2}+\frac{\sqrt{3}}{2}i}\right)=\text{Arg}\left(\left(\frac{1}{2}+\frac{\sqrt{3}}{2}i\right)^2\right)$ …. (2)

Use a CAS to solve equations (1) and (2) simultaneously under the restriction $a,b\in R^+$:

$a=1$ and $b=1$.

Note: The very convenient value $z=2$ does not give an equation that allows the unique solution for a and b.

Clear and valid method	**1 mark**
Answer: $a=1$ and $b=1$	**1 mark**

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Part d.:

$-\frac{\pi}{2} < \text{Arg}(z) < \frac{\pi}{2}$ for $z \in S = \{z: |z| = |z-1|, z \in C\}$

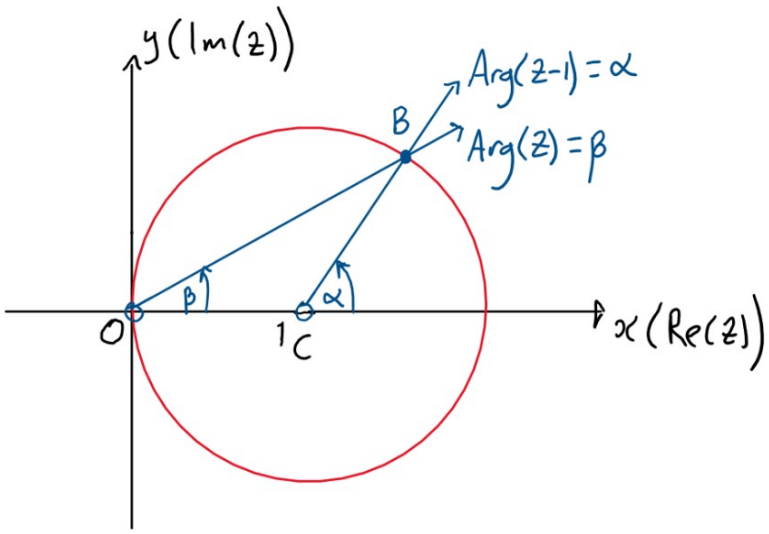
Note:

1. $z \to 0$ as $\text{Arg}(z) \to \pm\frac{\pi}{2}$ and $\text{Arg}(0)$ is not defined.
2. $\text{Arg}(z^2) = 2\text{Arg}(z)$ since $-\frac{\pi}{2} < \text{Arg}(z) < \frac{\pi}{2}$
3. $z = 0$ is the exceptional element of *S* that is not an element of $\{z: \text{Arg}(z-1) = \text{Arg}(z^2), z \in C\}$

(because $\text{Arg}(0^2)$ is not defined).

Case 1: $z \in S \setminus \{0\}$ and $0 \le \text{Arg}(z) < \frac{\pi}{2}$ (upper half of the circle).

Let $\text{Arg}(z-1) = \alpha$ and $\text{Arg}(z) = \beta$:



Proving $\text{Arg}(z-1) = \text{Arg}(z^2)$ is equivalent to proving $\alpha = 2\beta$:

$OC = CB$ ($= 1 =$ radius of circle)

therefore ΔOCB is isosceles

therefore $\angle BOC = \beta = \angle OBC$

therefore $\angle OCB = \pi - 2\beta$. …. (1)

But $\angle OCB = \pi - \alpha$. …. (2)

From equations (1) and (2): $\pi - 2\beta = \pi - \alpha \quad \Rightarrow 2\beta = \alpha$ ■

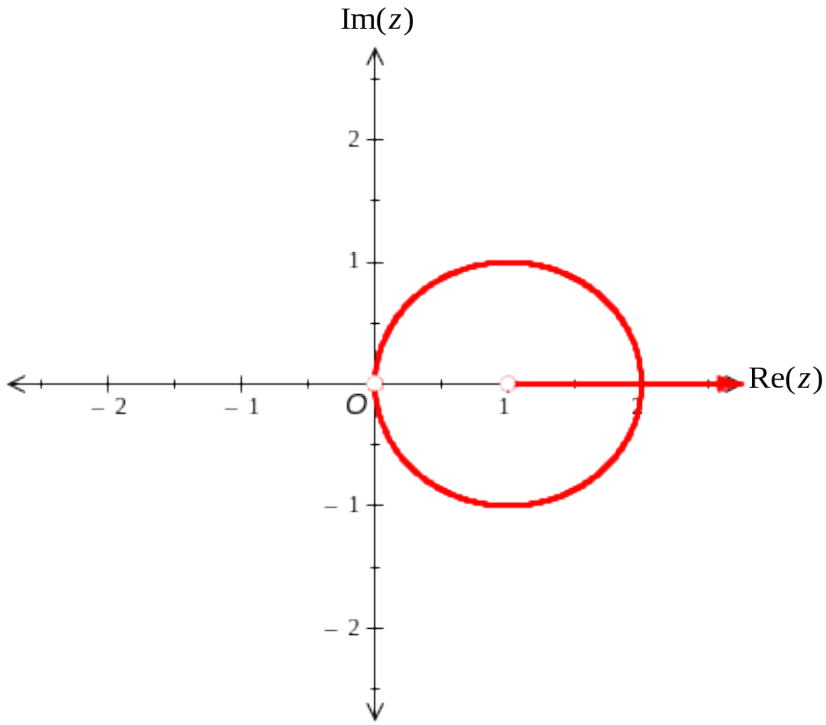
Case 2: $z \in S \setminus \{0\}$ and $-\frac{\pi}{2} < \text{Arg}(z) \leq 0$ (lower half of the circle).

The proof is identical to **Case 1** by symmetry.

$\text{Arg}(z^2) = 2\text{Arg}(z)$ since $-\frac{\pi}{2} < \text{Arg}(z) < \frac{\pi}{2}$ $z = 0$ is the exceptional element of S that is not an element of $\{z : \text{Arg}(z-1) = \text{Arg}(z^2),\ z \in C\}$	**1 mark**
Appropriate labelled diagram. Geometric proof.	**1 mark**

Part e.:

Answer:



Shape: Circle $\lvert z-1\rvert=1$ with 'hole' at $z=0$	**1 mark**
Shape: Ray with terminus at $z=1$	**1 mark**
'Hole' at terminus of ray.	**1 mark**

Calculations:

Shape:

- From **part d.** it is known that the circle $\lvert z-1\rvert=1$, $z\neq 0$, is part of the solution.
- From **part d.** it is seen that $\text{Arg}(z-1)$ and $\text{Arg}(z^2)$ are equal when z is real and $z>1$ ($\alpha=\beta=0$)

Alternatively, $\text{Arg}(z-1)=\text{Arg}(z^2)$ when $z>1$ can be seen by inspection. If z is real:

$z^2>0$ therefore $\text{Arg}(z^2)=0$

$z>1\Rightarrow z-1>0$ therefore $\text{Arg}(z-1)=0$ therefore $\text{Arg}(z-1)=\text{Arg}(z^2)$

$z<1\Rightarrow z-1<0$ therefore $\text{Arg}(z-1)=\pi$ therefore $\text{Arg}(z-1)\neq\text{Arg}(z^2)$

$\text{Arg}(z-1)$ is not defined for $z=1$ therefore $z=1$ is not a solution to $\text{Arg}(z-1)=\text{Arg}\left(z^2\right)$

Part f.:

Let $\text{Arg}(z-1)=\alpha$ and $\text{Arg}(z)=\beta$ where $\alpha=2\beta$ (from **Part d.**).

- $\text{Arg}\left(z^2\right)=2\text{Arg}(z)$

  since $z\in S$ and $-\frac{\pi}{2}<\text{Arg}(z)<\frac{\pi}{2}$ for *S*

  $=2\beta$

- $\text{Arg}\left(z^2-z\right)=\text{Arg}\left(z(z-1)\right)=\text{Arg}(z)+\text{Arg}(z-1)=\beta+\alpha=\beta+2\beta=3\beta$

provided

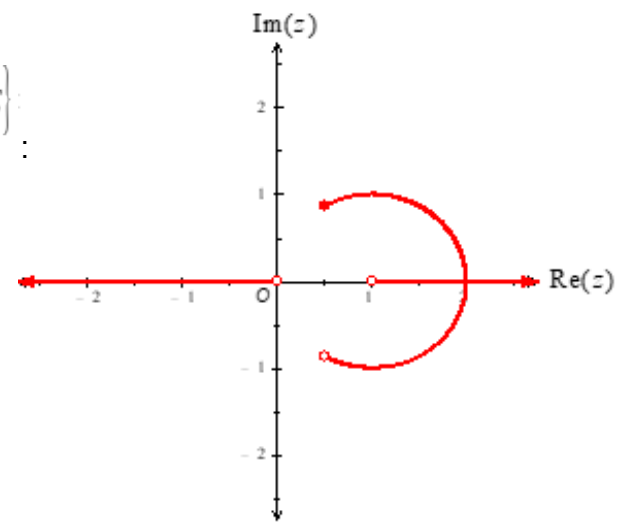
$-\pi<3\beta\le\pi \quad\Rightarrow -\frac{\pi}{3}<\beta\le\frac{\pi}{3}$

Then $3\text{Arg}\left(z^2\right)=2\text{Arg}\left(z^2-z\right) \quad\Rightarrow 3(2\beta)=2(3\beta)$ ✓

Recognition that $\text{Arg}\left(z^2-z\right)=\text{Arg}(z)+\text{Arg}(z-1)$ provided $-\pi<\text{Arg}(z)+\text{Arg}(z-1)\le\pi$	**1 mark**
Answer: $-\frac{\pi}{3}<\text{Arg}(z)\le\frac{\pi}{3}$	**1 mark**

Discussion:

Sketch of $\left\{z:\ 3\text{Arg}\left(z^2\right)=2\text{Arg}\left(z^2-z\right),\ z\in C\right\}$:



Calculations:

- It is known that the circle $|z-1|=1$ is part of the solution for $-\frac{\pi}{3} < \text{Arg}(z) \le \frac{\pi}{3}$
- It is seen by inspection that $3\text{Arg}(z^2)$ and $2\text{Arg}(z^2 - z)$ are equal when z is real and $z^2 > 0$ and $z^2 - z > 0$:

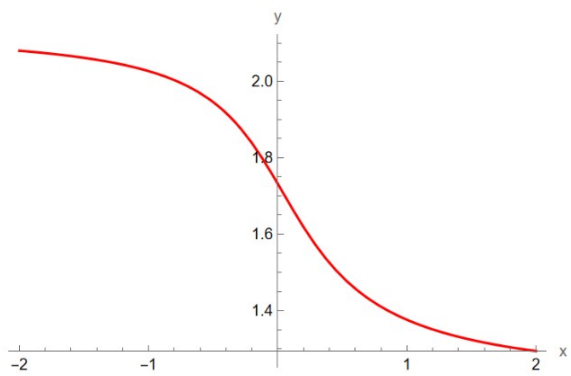
$z < 0$ or $z > 1$

Note: If z is real:

$z^2 > 0$ therefore $\text{Arg}(z^2) = 0$

- $z > 1 \Rightarrow z^2 - z > 0$ therefore $\text{Arg}(z^2 - z) = 0$ therefore $3\text{Arg}(z^2) = 2\text{Arg}(z^2 - z)$
- $0 < z < 1 \Rightarrow z^2 - z < 0$ therefore $\text{Arg}(z^2 - z) = \pi$ therefore $3\text{Arg}(z^2) \neq 2\text{Arg}(z^2 - z)$
- $z < -1 \Rightarrow z^2 - z > 0$ therefore $\text{Arg}(z^2 - z) = 0$ therefore $3\text{Arg}(z^2) = 2\text{Arg}(z^2 - z)$

Question 3



Part a.:

$V = \pi \int_0^1 \left(3 - \tan^{-1}(2x) \right)^2 dx$	**1 mark**

Answer: 7.2106 cubic units	**1 mark**

Part b.:

Integral terminals:

• $x = -1 \Rightarrow y = \sqrt{3 - \tan^{-1}(-2)}$. Accept $y = \sqrt{3 + \tan^{-1}(2)}$ • $x = 0 \Rightarrow y = \sqrt{3}$	**1 mark**

$S = 2\pi \int_{\sqrt{3}}^{\sqrt{3 - \tan^{-1}(-2)}} x\sqrt{1 + \left(\frac{dx}{dy}\right)^2}\, dy$ or $S = 2\pi \int_{\sqrt{3}}^{\sqrt{3 + \tan^{-1}(2)}} x\sqrt{1 + \left(\frac{dx}{dy}\right)^2}\, dy$ Accept $S = -2\pi \int_{\sqrt{3 - \tan^{-1}(-2)}}^{\sqrt{3}} x\sqrt{1 + \left(\frac{dx}{dy}\right)^2}\, dy$, $S = 2\pi \left\lvert \int_{\sqrt{3 - \tan^{-1}(-2)}}^{\sqrt{3}} x\sqrt{1 + \left(\frac{dx}{dy}\right)^2}\, dy \right\rvert$ *etc*	**1 mark** Order of terminals must be correct.

- Substitute $x = \frac{1}{2}\tan(3 - y^2)$:

Answer: 3.2253 square units.	**1 mark**

Part c.:

$L = \pi \int_{-1}^{1} \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx$	**1 mark**

Answer: 2.1233 units.	**1 mark**

Part d.:

Use $S = 2\pi \int_a^b y\sqrt{1+\left(\frac{dy}{dx}\right)^2}\,dx$:

- $S_f = 2\pi \int_a^b f(x)\sqrt{1+(f'(x))^2}\,dx$
- $S_g = 2\pi \int_a^b g(x)\sqrt{1+(g'(x))^2}\,dx$

$S_g = 2\pi \int_a^b (f(x)+k)\sqrt{1+(f'(x))^2}\,dx$	**1 mark**

$$= 2\pi \int_a^b f(x)\sqrt{1+(f'(x))^2}\,dx + 2\pi \int_a^b k\sqrt{1+(f'(x))^2}\,dx$$

$$= 2\pi \int_a^b f(x)\sqrt{1+(f'(x))^2}\,dx + 2\pi k \underbrace{\int_a^b \sqrt{1+(f'(x))^2}\,dx}_{L}$$

Answer: $S_g = S_f + 2\pi kL$	**1 mark**

Question 4

Part a.:

Require $\frac{dN}{dt} > 0$ at $t = 0$, that is, when $N = 150$: $2(150)\left(1 - \frac{150}{600}\right) - n > 0$

Answer: $0 < n < 225$	**1 mark**

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Part b.:

$$\frac{dN}{dt} = 2N\left(1 - \frac{N}{600}\right) - n \qquad \Rightarrow \frac{dt}{dN} = \frac{1}{2N\left(1 - \frac{N}{600}\right) - n} = \frac{1}{\frac{-N^2}{300} + 2N - n}$$

Require $\frac{-N^2}{300} + 2N - n$ to be a perfect square.	**1 mark**

Therefore $\Delta = (2)^2 - 4\left(\frac{-1}{300}\right)(-n) = 0$.

Answer: $n = 300$	**1 mark**

Check (use a CAS):

Solve $\frac{dN}{dt} = 2N\left(1 - \frac{N}{600}\right) - 300$ with initial condition $N(0) = 150$:

$$N = \frac{300(t-1)}{t-2} = 300 + \frac{300}{t-2}.$$

Observation: $N = 0$ when $t = 1$.

Part c.:

Use a CAS to solve $\frac{dN}{dt} = 2N\left(1 - \frac{N}{600}\right) - n$ with initial condition $N(0) = 150$:

$$N = 300 - 10\sqrt{3}\sqrt{n - 300}\tan\left(\frac{\sqrt{n - 300}}{10\sqrt{3}}t + \tan^{-1}\left(\frac{5\sqrt{3}}{\sqrt{n - 300}}\right)\right)$$

Note: Different CAS may give different but equivalent forms.

Substitute $t = 4$ and solve $N = 0$ for *n* (round to the nearest integer):

$300 - 10\sqrt{3}\sqrt{n - 300}\tan\left(\frac{4\sqrt{n - 300}}{10\sqrt{3}} + \tan^{-1}\left(\frac{5\sqrt{3}}{\sqrt{n - 300}}\right)\right) = 0$	**1 mark**

Answer: $n = 227$	**1 mark**

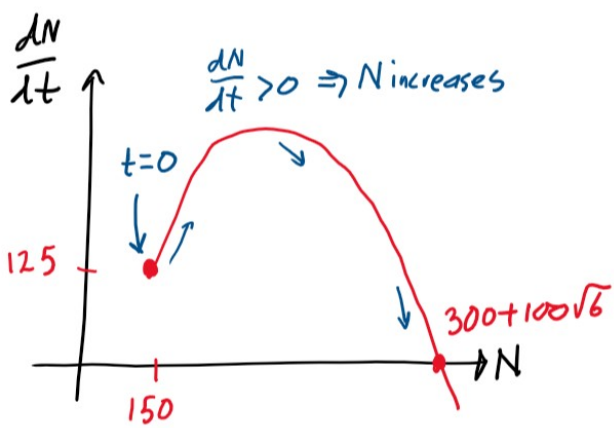
Check (use a CAS):

- Solve $\frac{dN}{dt} = 2N\left(1 - \frac{N}{600}\right) - 227$ with initial condition $N(0) = 150$
- Substitute $N = 0$ and solve for *t*: $t = 3.96966$ ✓

Part d. i.:

$\frac{dN}{dt} = 2N\left(1 - \frac{N}{600}\right) - 100$ with initial condition $N(0) = 150$

Phase diagram (plot of $\frac{dN}{dt}$ versus *N*)**:**



$\frac{dN}{dt} > 0$ at $t = 0$ ($N = 150$) therefore N initially increases.

$\frac{dN}{dt} > 0$ as N increases ($t > 0$) therefore N continues to increase.

$N \to 300 + 100\sqrt{6}$ as $\frac{dN}{dt} \to 0$.

Valid explanation that refers to a phase diagram. $N \to 300 + 100\sqrt{6}$	**1 mark**

Answer: 545	**1 mark**

Check (use a CAS):

- Solve $\frac{dN}{dt} = 2N\left(1 - \frac{N}{600}\right) - 100$ with initial condition $N(0) = 150$
- Calculate $\lim_{t \to +\infty} N(t)$: $300 + 100\sqrt{6}$

Part d. ii.:

$\frac{dN}{dt} = 2N\left(1 - \frac{N}{600}\right) - 100$ has a turning point at $N = 300$ therefore there is a point of inflection at $N = 300$

Answer: $N = 300$	**1 mark**

Check (use a CAS):

- Solve $\frac{dN}{dt} = 2N\left(1 - \frac{N}{600}\right) - 100$ with initial condition $N(0) = 150$
- Solve $\frac{d^2N}{dt^2} = 0$ for *t*: $t = \frac{1}{2}\frac{\sqrt{3}}{2}\log_e\left(\frac{4+\sqrt{6}}{4-\sqrt{6}}\right)$
- Substitute $t = \frac{1}{2}\frac{\sqrt{3}}{2}\log_e\left(\frac{4+\sqrt{6}}{4-\sqrt{6}}\right)$ into *N*: $N = 300$

Part e.:

- Use a CAS to solve $\frac{dN}{dt} = 2N\left(1 - \frac{N}{600}\right) - 100$ with initial condition $N(0) = 150$
- Substitute $N = 400$ and solve for t (round to two decimal places):

Answer: $t = 1.40$	**1 mark**

Question 5

Part a.:

If $\underset{\sim}{v}$ and $\underset{\sim}{w}$ are normal to each other then $\underset{\sim}{v} \cdot \underset{\sim}{w} = 0$:

$\left(-\underset{\sim}{i} - 2\underset{\sim}{j} + \underset{\sim}{k}\right) \cdot \left(3\underset{\sim}{i} - 2\underset{\sim}{j} - \underset{\sim}{k}\right) = -3 + 4 - 1 = 0$	**1 mark** Explicit calculation is required.

Part b.:

Method 1:

- A cartesian equation of the plane is $ax + by + cz = d$ where $\underset{\sim}{n} = a\underset{\sim}{i} + b\underset{\sim}{j} + c\underset{\sim}{k}$ is a vector normal to the plane.

  A normal vector to the plane is $\underset{\sim}{n} = \underset{\sim}{v} \times \underset{\sim}{w}$. From a CAS or 'by hand': $\underset{\sim}{v} \times \underset{\sim}{w} = 4\underset{\sim}{i} + 2\underset{\sim}{j} + 8\underset{\sim}{k}$

Normal vector: $4\underset{\sim}{i} + 2\underset{\sim}{j} + 8\underset{\sim}{k}$	**1 mark**

- Therefore $4x + 2y + 8z = d$

Substitute the point $(2, -2, -1)$ into $4x + 2y + 8z = d$ and solve for d: $d = -4$

Therefore $4x + 2y + 8z = -4$

• Explicit solution for *d* using the point $(2, -2, -1)$	**1 mark**

• **Answer:** $2x + y + 4z = -2$	Accept all equivalent answers.

Method 2:

- $(2, -2, -1)$ is a point contained in the plane Π and the vectors $\underset{\sim}{v}$ and $\underset{\sim}{w}$ are parallel to Π

Therefore a vector equation of Π is $\underset{\sim}{r}_{\Pi} = 2\underset{\sim}{i} - 2\underset{\sim}{j} - \underset{\sim}{k} + \lambda \underset{\sim}{v} + \mu \underset{\sim}{w}$ where $\lambda \in R$ and $\mu \in R$

Therefore a set of parametric equations defining Π is

$x = 2 - \lambda + 3\mu$ …. (1) $y = -2 - 2\lambda - 2\mu$ …. (2) $z = -1 + \lambda - \mu$ …. (3)	**1 mark**

- Use a CAS to solve equations (1), (2) and (3) simultaneously for λ, μ and one of either x, y or z and simplify the 'cartesian' solution.

(This is equivalent to solving two of the equations simultaneously for λ and μ in terms of x, y and z and then substituting those solutions into the third equation).

Answer: $2x + y + 4z = -2$	**1 mark**

Part c.:

From the answer to **Part b.**, it is noted that the point $(4,\ 2,\ -3)$ is contained in the plane Π.

The vectors $\underset{\sim}{v}$ and $\underset{\sim}{w}$ are parallel to Π.

Therefore a vector equation of Π is

$\underset{\sim}{r}_{\Pi} = 4\underset{\sim}{i} + 2\underset{\sim}{j} - 3\underset{\sim}{k} + \lambda\underset{\sim}{v} + \mu\underset{\sim}{w}$ where $\lambda \in R$ and $\mu \in R$	**1 mark** • Symbols different to $\underset{\sim}{r}$ and t must be used in the vector equation. • Parameters must be explicitly defined.

Note:

- A symbol different to $\underset{\sim}{r}$ must be used in the vector equation of Π because $\underset{\sim}{r}$ is the position vector of the object.
- A symbol different to t (such as λ) must be used to represent a parameter in the vector equation of Π because t represents time in the position vector of the object.

Choosing $\lambda = \cos(2t)$ and $\mu = \sin(3t)$ in $\underset{\sim}{r}_{\Pi} = 4\underset{\sim}{i} + 2\underset{\sim}{j} - 3\underset{\sim}{k} + \lambda\underset{\sim}{v} + \mu\underset{\sim}{w}$ defines the position vector of the object.	**1 mark**

Therefore the points on the path followed by the object are a subset of the points contained in the plane.	

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Part d.:

- $\underset{\sim}{r}_{\Pi} = 4\underset{\sim}{i} + 2\underset{\sim}{j} - 3\underset{\sim}{k} + \cos(2t)|\underset{\sim}{v}|\hat{\underset{\sim}{v}} + \sin(3t)|\underset{\sim}{w}|\hat{\underset{\sim}{w}}$ (from **part c.**).

- Since $\underset{\sim}{v}$ and $\underset{\sim}{w}$ are perpendicular to each other (from **part a.**), the unit vectors $\hat{\underset{\sim}{v}}$ and $\hat{\underset{\sim}{w}}$ parallel to the plane Π are analogous to the unit vectors $\underset{\sim}{i}$ and $\underset{\sim}{j}$ parallel to the xy-plane.

- Therefore the parametric equations defining the path followed by the object in the plane Π are equivalent to the parametric equations

$$x = |\underset{\sim}{v}|\cos(2t) = \sqrt{6}\cos(2t), \qquad y = |\underset{\sim}{w}|\sin(3t) = \sqrt{14}\sin(3t)$$

in the xy-plane.

$x = \sqrt{6}\cos(2t), \qquad y = \sqrt{14}\sin(3t)$	**1 mark**

- Substitute $x = \sqrt{6}\cos(2t)$, $y = \sqrt{14}\cos(2t)$, $t_1 = 0$ and $t_2 = 1$ into the arc length parametric formula $\int_{t_1}^{t_2}\sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\,dt$ and evaluate using a CAS:

Answer: 8.24	**1 mark**

Part e.:

$\underset{\sim}{s} = \underset{\sim}{i} + \underset{\sim}{j} + 6\underset{\sim}{k} + t\left(-\underset{\sim}{i} + 6\underset{\sim}{j} - \underset{\sim}{k}\right)$ is the vector equation of a line.

By inspection a set of parametric equations defining the line is

$x = 1 - t \quad \Rightarrow t = -x + 1$ …. (1)

$y = 1 + 6t \quad \Rightarrow t = \dfrac{y-1}{6}$ …. (2)

$z = 6 - t \quad \Rightarrow t = -z + 6$ …. (3)

Equate equations (1), (2) and (3):

Answer: $-x + 1 = \dfrac{y-1}{6} = -z + 6$ Accept all equivalent forms.	**1 mark**

Part f.:

- It is prudent to check whether or not the line intersects the plane Π (if the line intersects the plane then the distance is equal to zero).

Method 1:

Substitute the parametric equations

$x = 1 - t$, $y = 1 + 6t$, $z = 6 - t$

of the line into the equation $2x + y + 4z = -2$ of Π (from **part b.**) and solve for t:

$2(1 - t) + (1 + 6t) + 4(6 - t) = -2 \quad \Rightarrow 27 = -2$

which is inconsistent. Therefore the line does not intersect Π.

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Method 2: Determine whether or not the line is parallel to Π.

A vector in the direction of the line is $-\underset{\sim}{i}+6\underset{\sim}{j}-\underset{\sim}{k}$ (by inspection) and a normal vector to the plane is $4\underset{\sim}{i}+2\underset{\sim}{j}+8\underset{\sim}{k}$ (from **part b.**).

$$\left(-\underset{\sim}{i}+6\underset{\sim}{j}-\underset{\sim}{k}\right)\cdot\left(4\underset{\sim}{i}+2\underset{\sim}{j}+8\underset{\sim}{k}\right)=0$$

therefore the line is parallel to the plane.

Note: The line does not lie in Π since any chosen point on the line does not satisfy the equation of the plane.

- Calculate the distance.

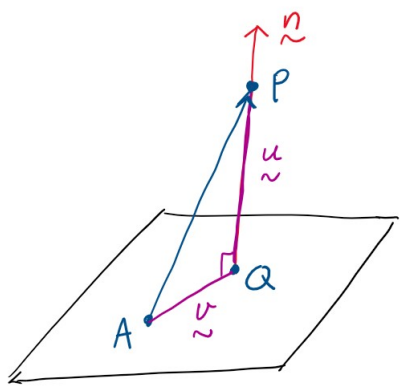
Note: Since the question is worth 2 marks, “appropriate working **must** be shown”.

Method	Example calculation
• Calculate the distance from Π of any chosen point P on the line.	Choose $P(1,\ 1,\ 6)$ (corresponding to $t=0$).
• Define the position vector of P.	$\overrightarrow{OP}=\underset{\sim}{i}+\underset{\sim}{j}+6\underset{\sim}{k}$.
• Get a normal vector $\underset{\sim}{n}$ to Π.	$\underset{\sim}{n}=4\underset{\sim}{i}+2\underset{\sim}{j}+8\underset{\sim}{k}$ (from **part b.**)
• Apply the distance formula $\dfrac{\left\lvert \underset{\sim}{n}\cdot\overrightarrow{OP}-d\right\rvert}{\lvert\underset{\sim}{n}\rvert}$ where d is found from the cartesian equation $ax+by+cz=d$ of Π.	From either a CAS or ‘by hand’: $\dfrac{\left\lvert\left(4\underset{\sim}{i}+2\underset{\sim}{j}+8\underset{\sim}{k}\right)\cdot\left(\underset{\sim}{i}+\underset{\sim}{j}+6\underset{\sim}{k}\right)-(-2)\right\rvert}{\lvert 4\underset{\sim}{i}+2\underset{\sim}{j}+8\underset{\sim}{k}\rvert}=\dfrac{4\sqrt{7}}{\sqrt{3}}$.

	Note: $d = -2$ comes from $2x + y + 4z = -2$ (from **part b.**)

Application of $\dfrac{\left\lvert \underset{\sim}{n} \cdot \overrightarrow{OP} - d \right\rvert}{\lvert \underset{\sim}{n} \rvert}$ **or any other valid method.**	**1 mark**
Answer: $\dfrac{4\sqrt{21}}{3}$	**1 mark**

Derivation of distance formula:



$|\overrightarrow{PQ}| = |$scalar resolute of $\overrightarrow{AP}$ in the direction of $\underset{\sim}{n}|$

$$= \frac{|\overrightarrow{AP} \cdot \underset{\sim}{n}|}{|\underset{\sim}{n}|} = \frac{|(-\overrightarrow{OA} + \overrightarrow{OP}) \cdot \underset{\sim}{n}|}{|\underset{\sim}{n}|}$$

$$= \frac{|\underset{\sim}{n} \cdot \overrightarrow{OP} - \underset{\sim}{n} \cdot \overrightarrow{OA}|}{|\underset{\sim}{n}|}$$

$$= \frac{|\underset{\sim}{n} \cdot \overrightarrow{OP} - d|}{|\underset{\sim}{n}|}$$

$\underset{\sim}{n} \cdot \overrightarrow{OA} = d$:

From the vector eq^n of a plane: $\underset{\sim}{r} \cdot \underset{\sim}{n} = \overrightarrow{OA} \cdot \underset{\sim}{n}$

$\Rightarrow ax + by + cz = \overrightarrow{OA} \cdot \underset{\sim}{n}$

$\Rightarrow d = \overrightarrow{OA} \cdot \underset{\sim}{n}$

Question 6

Part a.:

- Let X be the random variable "*Amount of manure (kg) in a small bag*":

$X \sim \text{Normal}(\mu_X,\ \sigma_X = 0.75)$

- Let the random variable $W = X_1 + X_2 + X_3$

where X_1, X_2 and X_3 are independent copies of X:

$\mu_W = \mu_{X_1} + \mu_{X_2} + \mu_{X_3} = 3\mu_X$

$\sigma_W^2 = 1^2\sigma_{X_1}^2 + 1^2\sigma_{X_2}^2 + 1^2\sigma_{X_3}^2 = 3\sigma_X^2 = 3(0.75)^2$ therefore $\sigma_W = 0.75\sqrt{3}$

W follows a normal distribution since X_1, X_2 and X_3 are independent normal random variables.

Definition of appropriate random variables. $W \sim \text{Normal}(3\mu_X,\ \sigma_W = 0.75\sqrt{3})$	**1 mark**

- $\Pr(W > 46) = 0.6239$.

$Z = \dfrac{W - \mu_W}{\sigma_W} \Rightarrow z = \dfrac{46 - 3\mu_X}{0.75\sqrt{3}}$ where $\Pr(Z > z) = 0.6239 \Rightarrow z = -0.31574$

$\dfrac{46 - 3\mu_X}{0.75\sqrt{3}} = -0.31574$	**1 mark** Give 1 mark if z-value is wrong but do not give the answer mark.

- Use a CAS to solve for μ_X: $\mu_X \approx 15.4701$

Answer: $\mu_X = 15.47$	**1 mark** There must be evidence that Z was used.

Rounding check: $\mu_X = 15.47 \Rightarrow \Pr(W > 46) = 0.62385 \approx 0.6239$ ✓

Part b.:

The statistical test is applied at the 3% level of significance therefore the probability of a type I error is $\frac{3}{100}=0.03$

Answers: • $H_0: \mu=25$ • $H_1: \mu\neq 25$ • 0.03	**1 mark**

Part c.:

- $\bar{x}=\frac{807.4}{32}=25.23125$
- $p=2\Pr\left(\bar{X}>25.23125 \middle| H_0 \text{ true}\right)$
- Two-sided test therefore .

Under H_0: $\bar{X}$ ~ Normal $\left(\mu_{\bar{X}}=\mu_X=25,\ \sigma_{\bar{X}}=\frac{\sigma_X}{\sqrt{n}}=\frac{1}{\sqrt{32}}\right)$

Therefore:

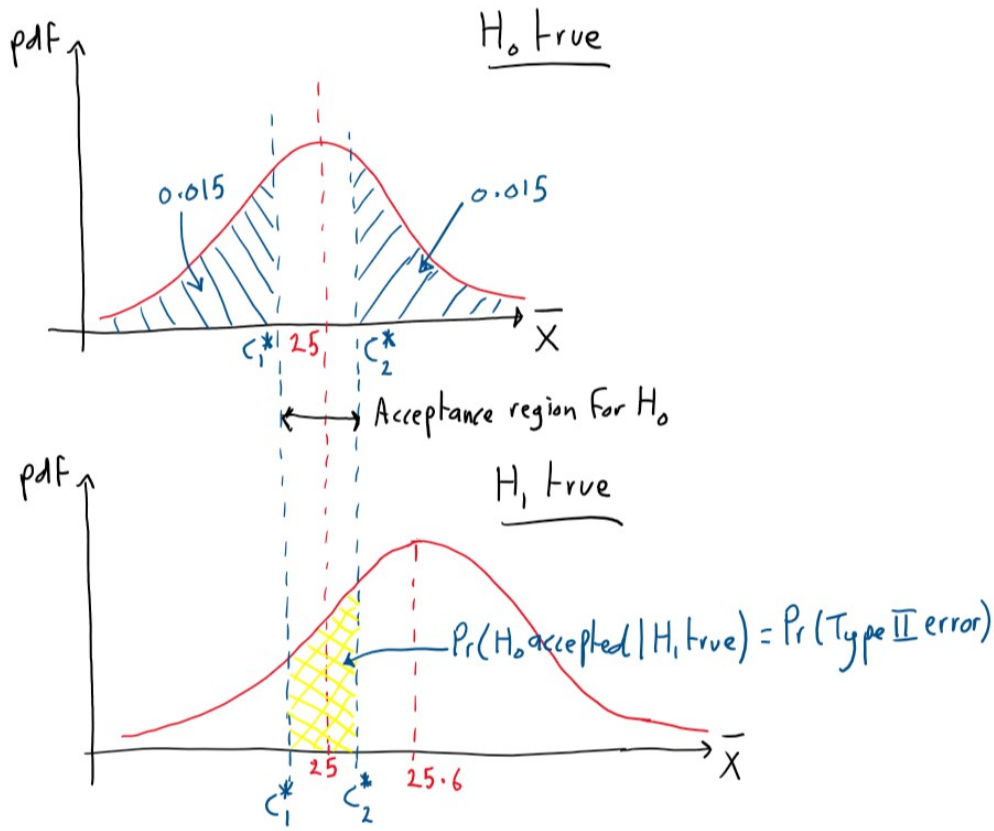
$p=2\Pr\left(\bar{X}>25.23125 \middle| \bar{X}\sim \text{Normal}\left(\mu_{\bar{X}}=25,\ \sigma_{\bar{X}}=\frac{1}{\sqrt{32}}\right)\right)$ Accept all equivalent statements such as $p=\Pr\left(\left\|\bar{X}-25\right\|>0.23125 \middle| \bar{X}\sim \text{Normal}\left(\mu_{\bar{X}}=25,\ \sigma_{\bar{X}}=\frac{1}{\sqrt{32}}\right)\right)$	**1 mark**

- From a CAS:

Answer: 0.0954	**1 mark**

Part d.:

- Definition: A type II error is the failure to reject H_0 when H_0 is false.
- Critical values C_1^* and C_2^* of $\overline{X}$ (use a CAS):



Note 1:

C_1^* and C_2^* are symmetric about $\mu_{\overline{X}} = 25$ by symmetry of the normal distribution.

Note 2:

The interval $(C_1^*,\ C_2^*)$ is **not** a confidence interval. It is the acceptance region for H_0

- H_0 is accepted when $\overline{x} \in (C_1^*,\ C_2^*)$

Option 1:

$\Pr(\overline{X} > C_2^*) = \frac{1}{2}(0.03) = 0.015 \Rightarrow C_2^* = 25.3836214$. $\Pr(\overline{X} < C_1^*) = 0.015 \Rightarrow C_1^* = 24.61637859$.

Option 2:

$\Pr\left(25-k<\overline{X}<25+k\right)=0.97 \Rightarrow k=0.383621$ $C_1^*=25-k$ and $C_2^*=25+k$

A correct critical value: Either $C_1^* = 24.61637859$ or $C_2^* = 25.3836214$	**1 mark** More than three decimal place accuracy is required (to avoid rounding error in final answer)
Statement of probability of type II error using critical values: $\Pr\left(C_1^* < \overline{X} < C_2^* \middle\| \mu_{\overline{X}} = 25.6\right)$	**1 mark**
Answer: 0.110	**1 mark**